Lecture Five and Six

Creative Thinking for AI and IoT Innovations:

Agent → Intelligent Agent → Creative Agent

1. Choosing Creativity and Thinking
2. Lateral – Decoding thinking, Psychological and Historical creativity
3. Conceptual Spaces: Cognition, Thinking, Agents and Machines converging to creative outcomes

Parag Kulkarni, PhD, DSc

Projects/Assignments

- Projects
  - Part 1: Clear Project Statement, Database identification, Baselines if any, implementation strategy, Algorithm
  - Part 2: Implementation, working prototype, intermediate results if any, conclusion and findings (Brief Presentation and QA)

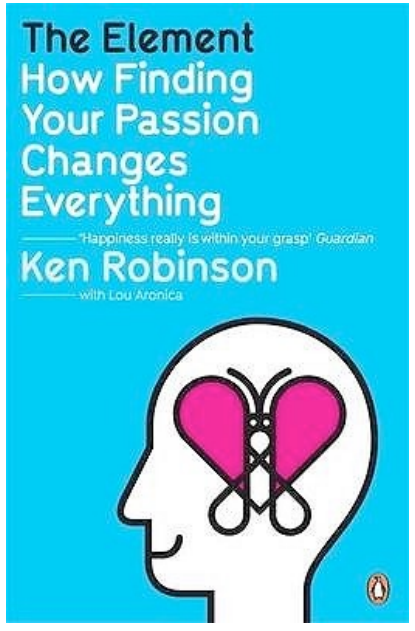
Quick Recap

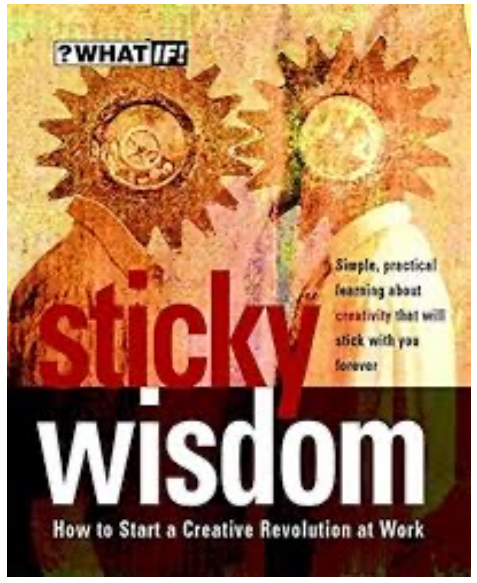
Opportunities…

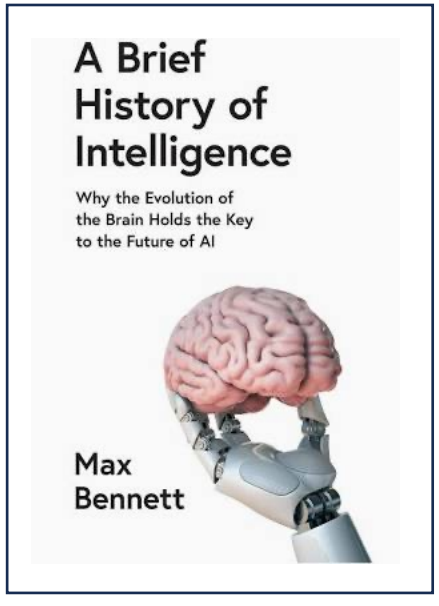
Creative Machine Learning…
(Creativity + Emotions + Consciousness + ?) = Human

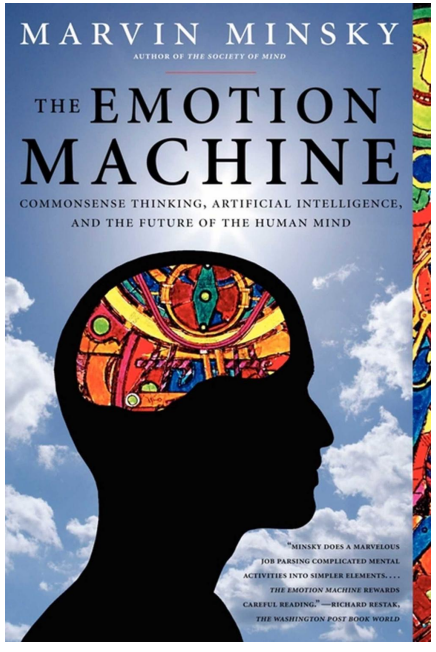
Intelligence is being human?

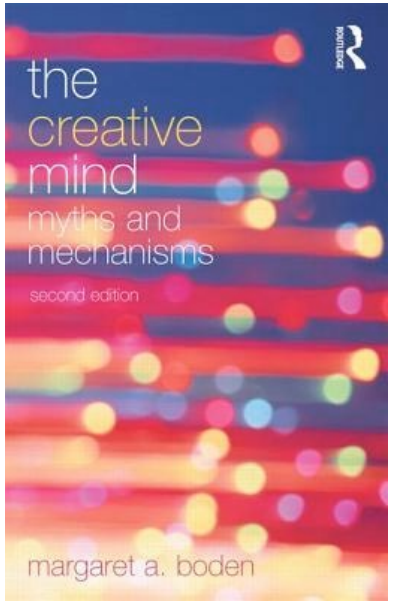
- Keep Learning!!!!!
- Historical creativity and Psychological Creativity
- Evolving systems

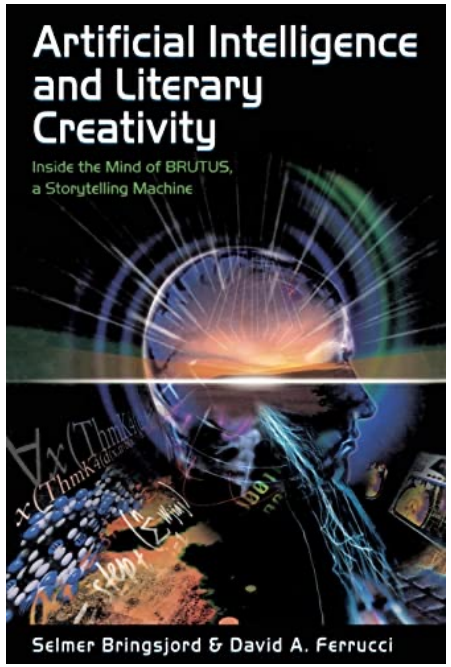


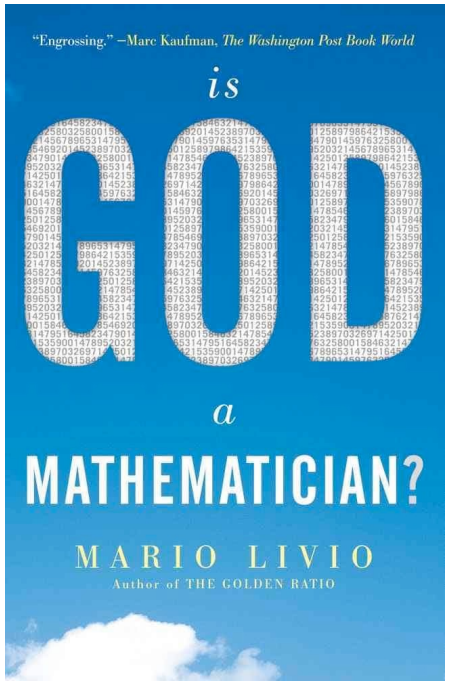




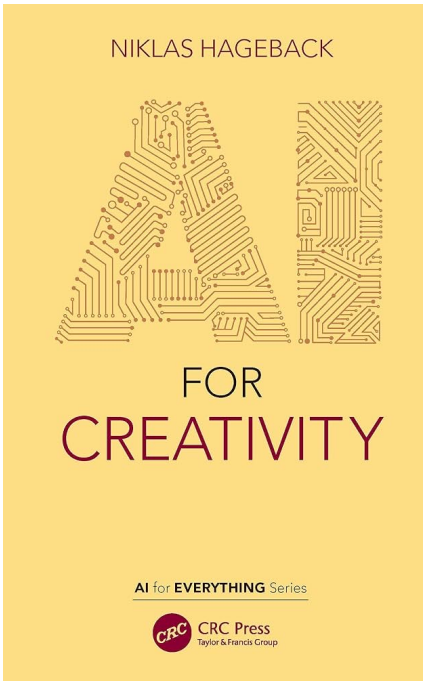


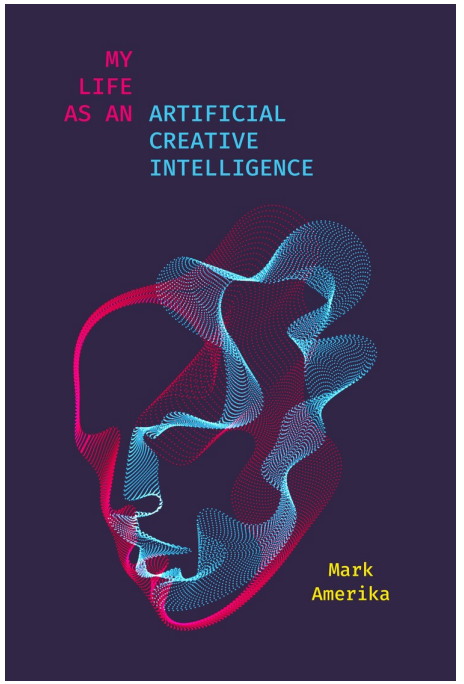












Lateral and creative thinking

Ideas to choice Opportunities...

GENERATION DECODED

How are novel ideas produced at all? Combination, exploration, transformation of conceptual spaces.

MORE THAN SELECTION

Given many candidates, which are worth having — and how do we compute that against a person's actual criteria rather than a generic average?

Choices, not answers

The system reveals a ranked set of genuine options with visible reasoning, rather than issuing one verdict the person can only accept or reject.

Trade-offs made visible

Every selection embeds a weighting of competing goods. Exposing the weighting lets the human disagree with it — which is where their judgement enters.

Human judgement structured, not replaced

The system organises the decision; the person makes it. In creative work this is not a limitation but the correct division of labour.

The Key Concept: Artificial Creative Intelligence

Computational Creativity	The established research field — systems evaluated for creative output against defined criteria
Creative AI	The applied framing — AI as a tool deployed within existing creative industries
Artificial Creative Intelligence	Creativity as a hybrid, distributed process moving across the human–nonhuman spectrum, belonging fully to neither party

counters the traditional creative virtues — authenticity, motivation, empowerment, intentionality — with a different set: intuition, spontaneity, remix. The shift is from creativity as expression of a self to creativity as a process passing through one.

The Key Concept: Lingual spontaneity and the Hybrid Mind (Mark)

*The **surrealists** (**Surrealists** were 20th-century artists and writers who sought to unlock the power of the unconscious mind and bypass rational thoughts) called it psychic automatism: writing without conscious control. Mark Amerika asks whether a generative transformer is doing something structurally similar — and whether AI might develop artificia*

Kerouac modelled his writing after jazz greats like Charlie Parker, treating the typewriter like a saxophone where the writer "blows" unedited thoughts onto the page.

1 The lineage he draws on

Surrealist automatism: · Kerouac's spontaneous prose · jazz improvisation · the Gysin–Burroughs cut-up method. All are techniques for producing text while 'losing consciousness'.

Mind, remixed

Collaboration generates a third authorial presence belonging to neither party. Remixing for the Hybrid Mind.

3 The Hybrid Mind

Not the human author, not the model, but the entity that emerges from their exchange

4 The forward-looking claim

If human intuition is a trained, automatic process, then artificial intuition is not a category error but an engineering horizon. (from Artificial Creative Intelligence Book)

The Key Concept: Terms

Artificial Creative Intelligence (ACI)

Creativity as a hybrid process distributed across human and nonhuman agents

Meta Remix Engine

The artist's own unconscious recombinatory apparatus

Psychic automatism

Surrealist term: composition without conscious control

Cosmotechnical persona

A style-identity understood as technically and cosmologically situated

Onto-operational presence

A mode of being constituted by operation rather than prior substance

Hybrid Mind

The third authorial entity emerging from human–machine exchange

Metamediumystic (Intuitive fluid way to operate)

His coinage for the mystical dimension of working through media

Séance of writing

Composition as summoning (Ordering to appear) — mediating presences not one's own

Exercise

Choose a creative artefact you admire — a piece of music, a product, a proof, a building. Classify the creativity involved using Boden's three forms, and defend the classification.

1 Name the conceptual space

What were the rules, conventions and constraints in force before this artefact existed? Be specific — vagueness here makes the rest impossible.

2 Classify the move

Was it a combination, an exploration, or a transformation? Most interesting cases are contested — argue for one and note what would count against you.

3 Locate the value judgement

Who decided it was good, on what basis, and when? Was the value recognised immediately or only later?

4 Ask the machine question

Which part could a system have done? Which part could it not? Be precise about why — this is the exercise's actual purpose.

Creative Computing: Search and Creativity

- Human Learning and Machine Learning: Unfolding from Creativity Perspective
- Looking at Exploratory Paradigms of Explainability in Creative Computing

https://philpapers.org/rec/KULHLA

https://www.taylorfrancis.com/chapters/edit/10.1201/9781003442509-2/

Creativity

- P creativity
- H Creativity
- S Creativity
- T Creativity

TURING TEST



Machine Response



Human Responder



Interrogator / investigator

CREATIVITY TEST



Machine Response



Human Responder



Interrogator / investigator

Creativity evaluator

Some responses may be creative!

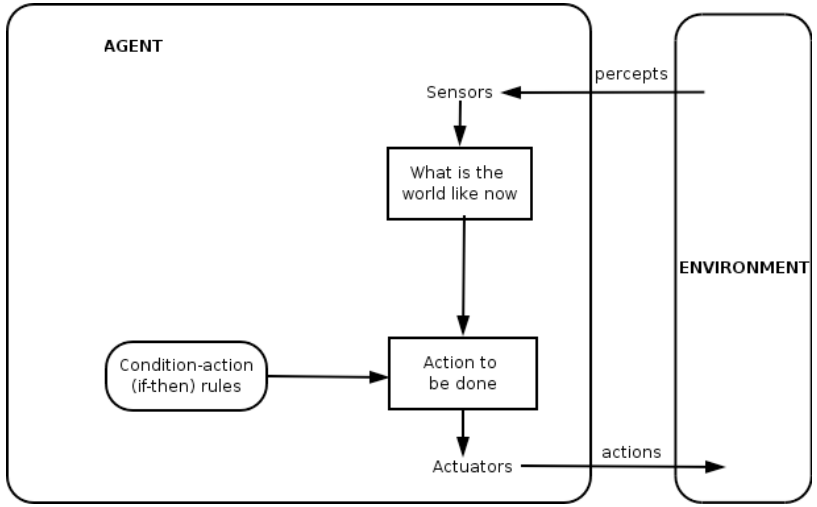
Cannot detect: Who is the originator of creative response while some of them are also coming from machine

Different approaches and views of creativity:

- Nomothetic versus idiographic view: The nomothetic approach focuses on universals of creativity while the idiographic focuses on individual differences and uniqueness of creative individuals
- Mental versus social phenomenon: Creativity is seen as a mental or psychological phenomenon for the generation of new ideas or as the result of an audience's appreciation of a novel idea.
- Improbabilist versus impossibilist approach: The improbabilist approaches see creativity as a novel combination of familiar ideas while the impossibilists consider creativity as new ideas never appeared before. This is possible thanks to a transformation of the conceptual space.
- Rational versus irrational process: The rational approaches include logic and evolution. Logic sees creativity in processes like abduction (guessing a conclusion given some premises), eduction – inferring (relate properties of different objects), deduction and induction (relate properties from general to specific and vice versa). Evolutionary approaches instead see creativity as an adaptation to the environment motivated by problem solving while the irrational approach focuses on the emotional part of creativity.

Agent

- Perceiving its environment through sensors and action on environment through actuators
- Percept sequence
- Agent Function
- Rationality – Rational agent



PEAS

- PEAS: Performance measure, Environment, Actuators, Sensors
- Must first specify the setting for intelligent agent design
- For creative agent what could be PEAS
  - +Performance measure
  - +Environment
  - +Actuators
  - +Sensors



Creative choosing

- Generation of multiple options
- The options are created through randomness but with objective
- These options are presented to a program
- A creative choosing will try to choose the best one with reference to various creativity evaluation mechanisms
- The ultimate choices will be presented to user for creation of final solution

Looking at it from Heuristic search perspective

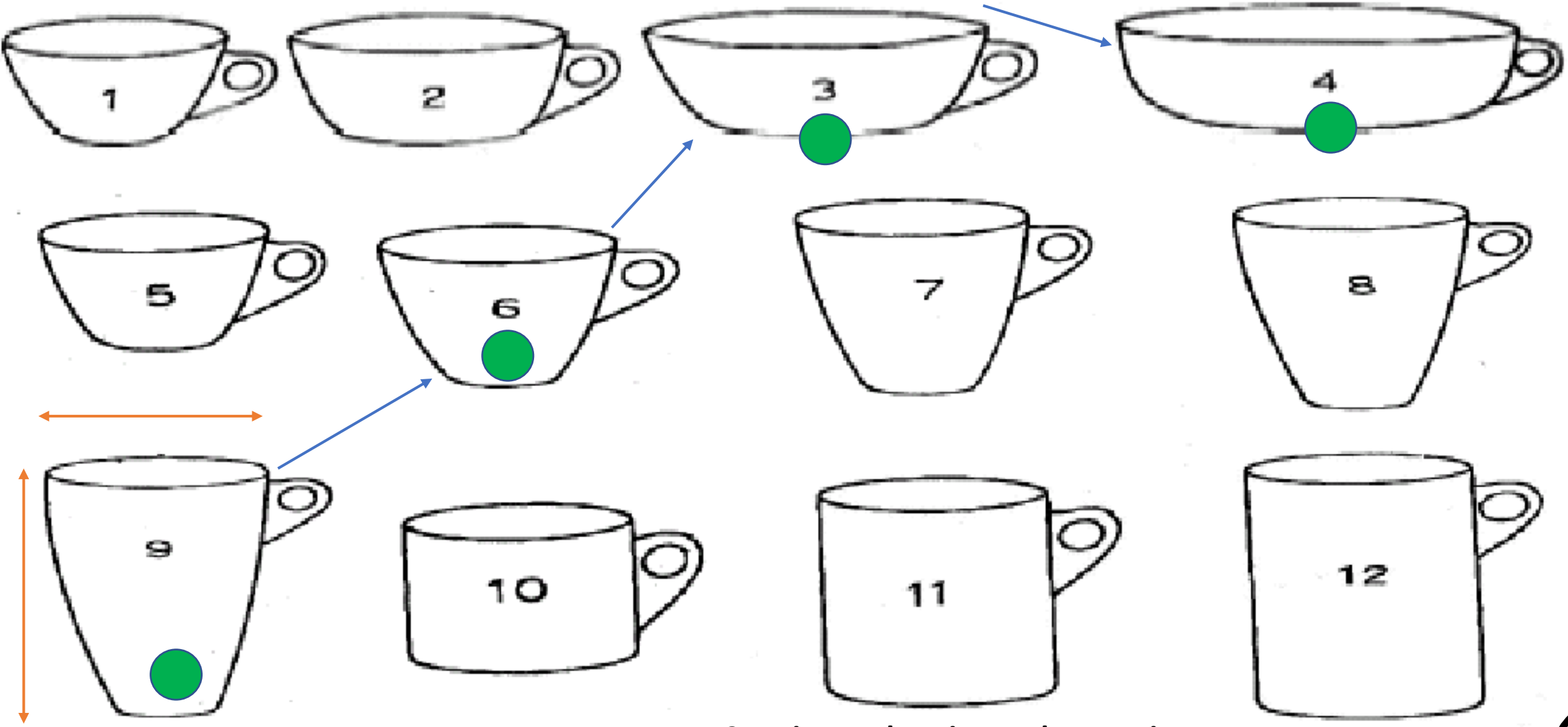
Modeling and evaluation of creativity process

- Expansion
- Exploration
- Analogy
- AUT – Alternate usage test
- Torrance Tests of Creative Thinking

What could be?

- Assigning properties to the object
- Properties → Concept → Concept expansion
- Goal State
- Start State
- Action Sequence
- Path Cost

Search/Exploration for the creative object -!
What could be heuristic function?

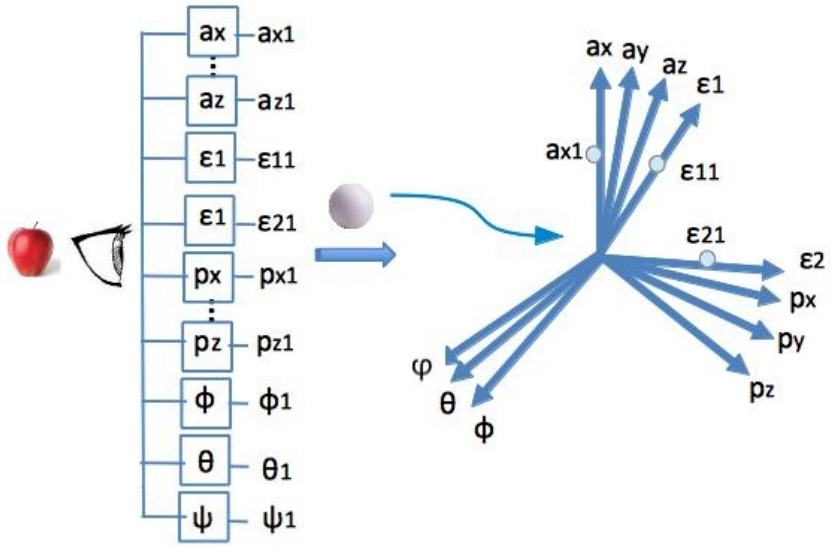


Creative Exploration and expansion

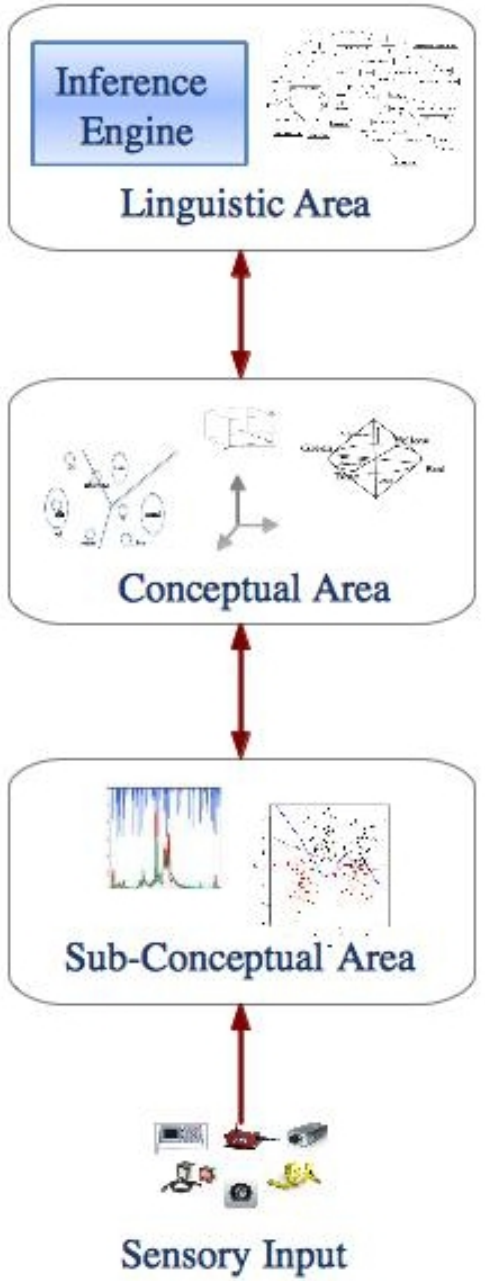
Creativity Traversals

- Looking for the best neighboring state
- Traversal to the most creative neighbor
- Creative state and creative paths
- Creative state and cumulative creative efforts
- New creative values for each state (C_t)
- Think of multiple starting state (Multi state search)
- Short list a few actions – short list few initial states – imagine about possible transformations

Conceptual Spaces



A **conceptual space** is a geometric structure that represents a number of quality dimensions, which denote basic features by which concepts and objects can be compared, such as weight, color, taste, temperature, pitch, and the three ordinary spatial dimensions.



Ref: conceptual creativity

Conceptual spaces

- Information represented by geometric structures
  - objects are points
  - properties and relations are regions
- Similarity is measured by the distance between points or regions in space
- Conceptual space defined by quality dimensions

Properties and Concepts

- Properties are distinct from concepts in conceptual model
  - Not covered by symbolic and connectionist representations
  - Property is a subset of concept
- Property - based on a single domain
- Concept - based on one or more domains
- Semantically, properties represent adjectives and concepts represent nouns

Concepts

- Concepts change depending the relevance of different domains for the given problem
- ex. “Apple” concept:

Domain	***Region***
Color	Red-yellow-green
Shape	Roundish (cycloid)
Texture	Smooth
Taste	Regions of the sweet and sour dimensions
Fruit	Region defined by specification of various characteristics
Nutrition	Values of sugar content, vitamins, fiber, etc.

- Relative weightings of domains are determined by the context

Conceptual Spaces by Peter Gardenfords (MIT Press)

Concept combinations

- property-concept combinations
  - XY where X is property, Y is concept
  - compatible – intersection of concepts
  - incompatible – for incompatible regions, the region of X overrules the region of Y

For example, pink elephant:

- X = pink, Y = elephant
- region of color domain for elephant (the gray area) is overruled by pink color domain

What about?

- Conceptual contradictions
- Conceptual mutations
- Conceptual cross overs
- Conceptual hierarchy
- Conceptual Transformation

Releasing the boundaries of compatibility

- The "boundaries of compatibility in creativity" refers to the idea that creativity thrives within constraints, rather than being limited by them, as structure and limitations can provide focus and foster innovation. These boundaries can be internal, like self-imposed project parameters, or external, such as client briefs, budgets, or even time constraints. When boundaries are clear and respected, they provide a stable foundation for creative exploration, prevent overwhelm, and guide collaborative efforts, leading to more focused and impactful work.
- Releasing these boundaries can lead to new conceptual definitions and formation of new spaces

Concept combinations

- Shifts caused by X of a region in Y in one domain can cause shifts for regions in Y in other domains because of correlations
- Example: brown apple. Brown modifies color domain of apple and also the texture domain, Clustered Apple ?





Stretching boundaries

- Obvious → Creative → ?
- Expanding a concept with another concept
- Conceptual spaces are formal frameworks that represent concepts as regions in a multidimensional space, defined by various quality dimensions like hue, saturation, or pitch. Boundaries in conceptual spaces delineate these regions, with vagueness often represented by thick or topologically open sets, while exploration involves discovering and understanding these structures, often through creative processes that push against established boundaries to generate new ideas and knowledge.

Primary and Secondary conceptual spaces

- Primary conceptual spaces represent direct, fundamental aspects of information like physical qualities, serving as the "building blocks" of understanding, while secondary conceptual spaces are derived from these, often representing more complex, abstract, or symbolic concepts that are built upon the primary ones.

Extensions and Expansions

- Conceptual blending good for language, but can be extended beyond?
- Such as:
  - Computational narrative
  - User-interface design
  - Gaming
  - Database Integration, Querying

Concept transformation

- Concept transformation in creativity is the process of reshaping, evolving, or restructuring existing ideas, elements, or conceptual frameworks to create something new and original. This goes beyond simply combining known ideas (combinatorial creativity) to include modifying the fundamental rules and boundaries (enabling constraints) of a domain itself, often leading to more radical and impactful innovations.

Types of Thinking

- Perpetual or concrete
- Conceptual or abstract
- Logical with reasoning
- Creative
- Problem based – problem solving oriented
- Convergent Vs Divergent

Convergent Vs Divergent thinking

- **Convergent Thinking** focuses on attaining solution of the problem – it aims to converge to solution and hence many times constrained
- In contrast, **divergent thinking** is about looking in different directions – opening mind and going beyond boundaries – it also about trying out multiple solutions for given problem

Thought Experiments – Creative Learning with limited Success

- Taking contrary view
- Thinking one subject in terms of another subject
- Taking multiple perspectives
- Hunting opportunities to think differently
- Resulting new relevant(?) combinations using existing ones

To understand creativity!

- We may need to decode the thought process of creators

Crab-inspired toys



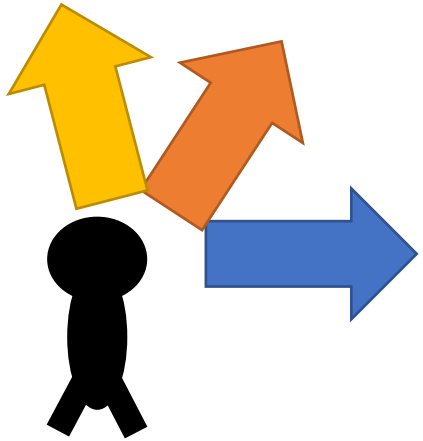
Combine elephant & butterfly

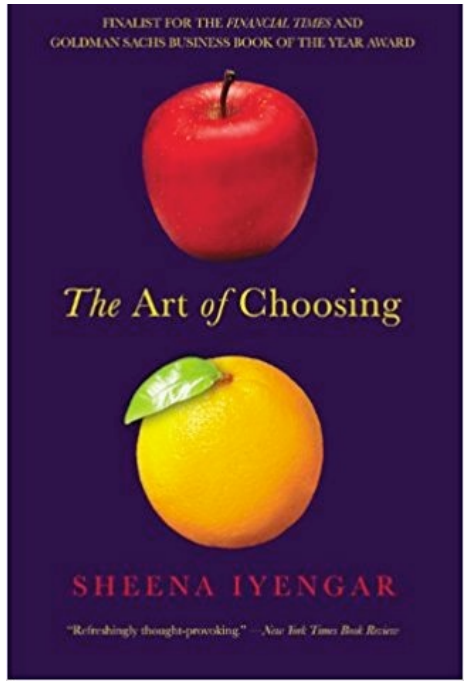


https://hbr.org/2023/07/how-generative-ai-can-augment-human-creativity

Choosing and creativity

Choice Engineering





Creative computing

Goal of Computational Creativity: To model, simulate or replicate creativity using a computer.

- How machines can help humans to exhibit creative behavior?
- How machines can learn from human creativity to exhibit human like creative behavior?
- How machines and humans could co-create the creative actions/products or artefacts.

By

- (Path correction/route mapping through suggestions, options, directives, information pointers or by providing platform)
- (By understanding process of creativity by creative individuals, going beyond creating random options, coming up with new + non-obvious + useful in the given context)
- (Understand coworkers' behaviors, understanding complementing and supporting sizable component, aiding with independent creation of some sizable portions)

Computational Creativity

- *Creativity is a process involving the generation of new ideas or concepts, or new associations between existing ideas or concepts*

  -- Wikipedia.

- Humans are creative. Ability to think out of the box.
- Goal of Computational Creativity: To model, simulate or replicate creativity using a computer.

Why Creativity?

- To construct a program/computer capable of human-level creativity.
- To better understand human creativity and to formulate an algorithmic perspective on creative behavior in humans.
- To design programs that can enhance human creativity without necessarily being creative themselves.

1
Begin with a Problem...

2
Identify Learning Opportunities...

3
ML Model is the key...

4
Model to value Creation...

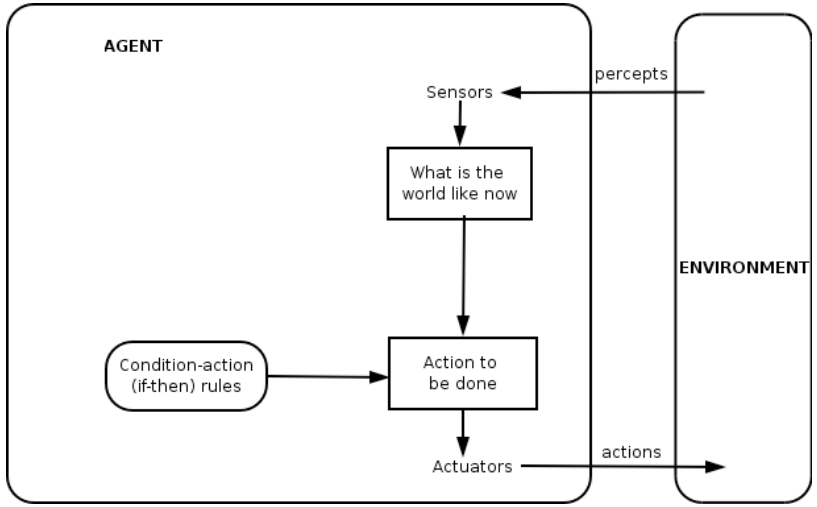
5
Generalization and Continuous Learning...

6
Generative, Deep and Exploratory...



Agent

- Perceiving its environment through sensors and action on environment through actuators
- Percept sequence
- Agent Function
- Rationality – Rational agent



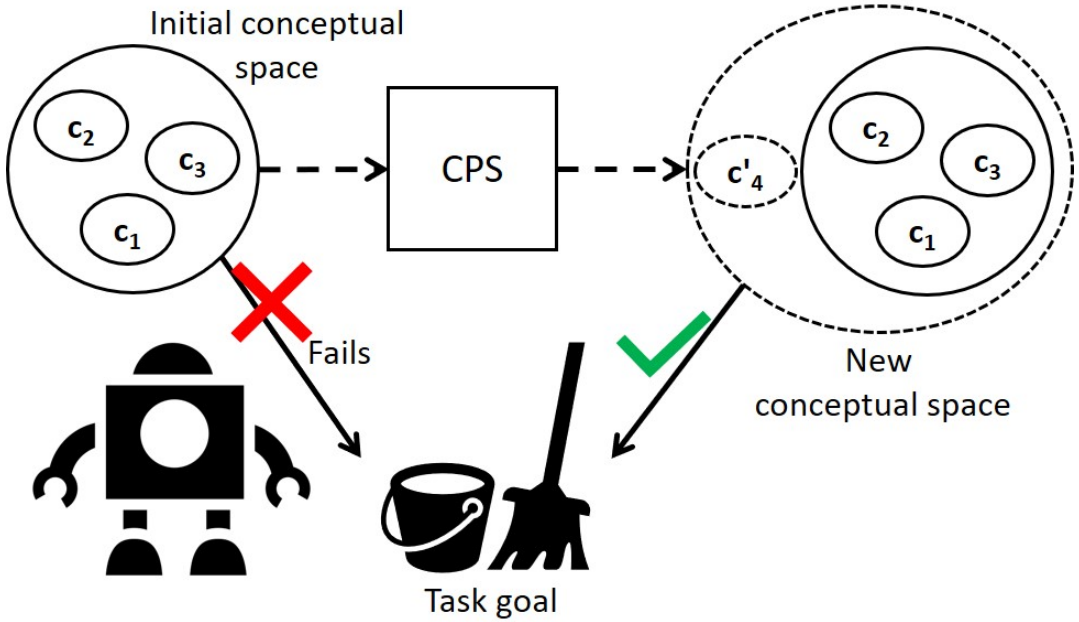
PEAS

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Creative Problem Solving In AI Agents

- Initial Conceptual space of agent is insufficient to solve problem

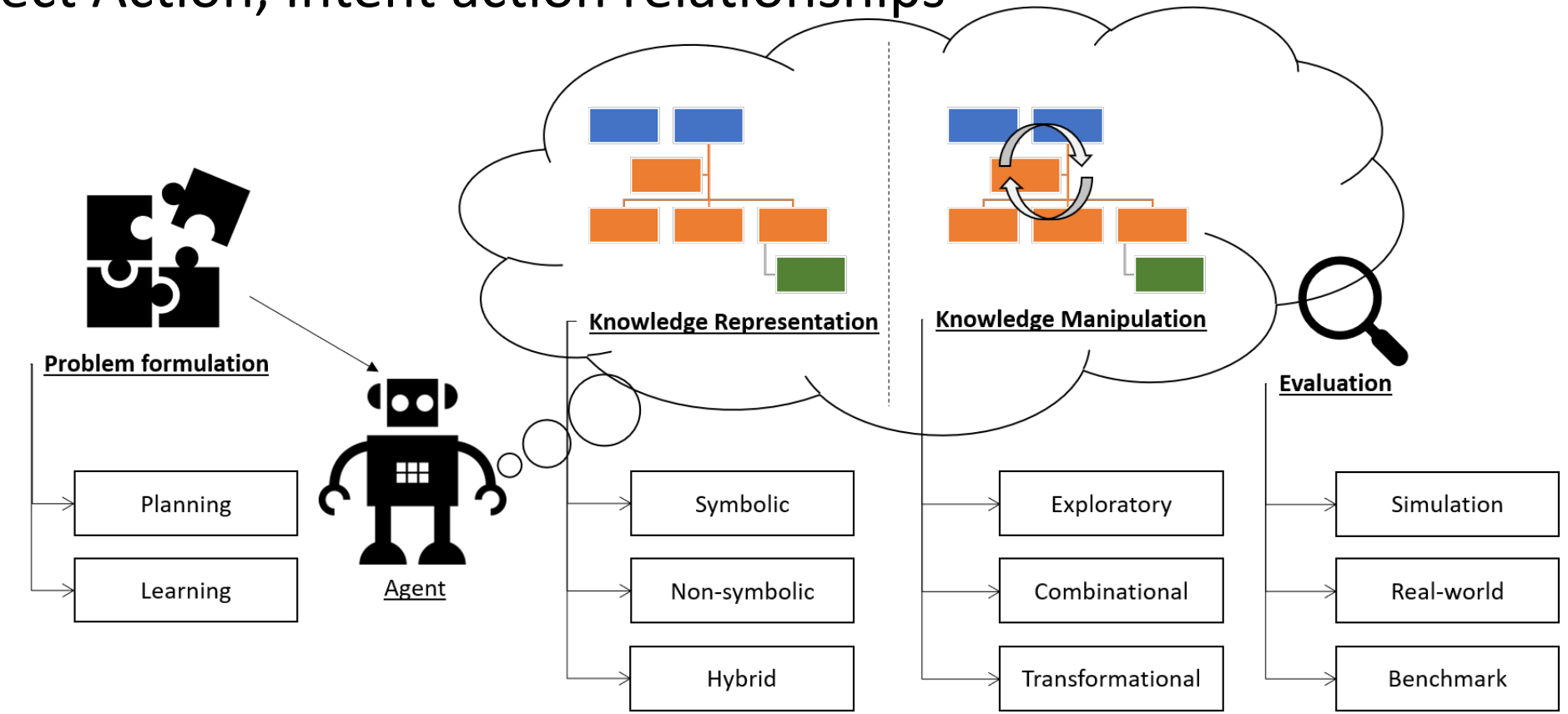


- How to Expand it?
- Context?
- Additional tools?
- Creation of…

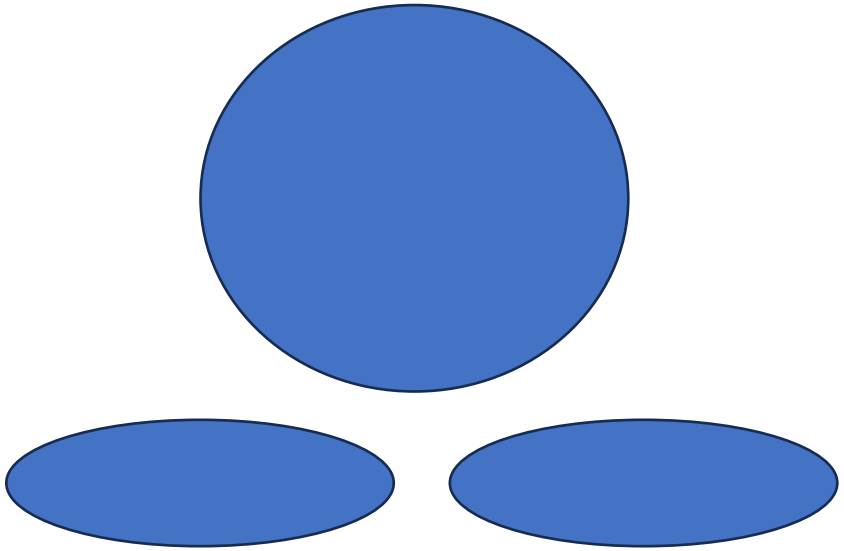
Ref:https://mulip.cs.tufts.edu/papers/gizzi_jair_2022.pdf

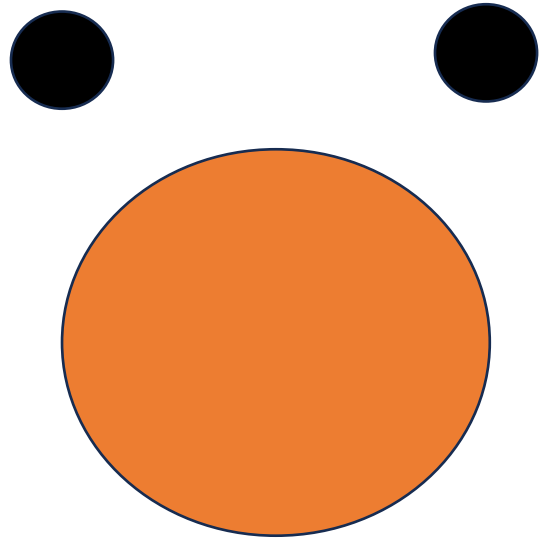
Creative Problem-Solving Framework (policy optimization)

- Object Action, Intent action relationships



Creativity





- Creative Agents Creative agents are poised to revolutionize content creation by generating and editing images, videos, audio, and copy while adhering to brand guidelines. Google introduced AI-powered Workspace apps, including auto-generated videos, as part of this initiative. With creative agents, businesses of all sizes will have access to advanced content creation capabilities, democratizing the creative process.

Creative agent

- Repurpose
- Complete the story differently
- Story formation
- Associate words, optimal word, repurpose words
- Building blocks to design different shapes
- Incremental design – evolutionary design
- Exploratory design
- Substitute, eliminate, what is minimal, what is optimal, adaptation, combine, expand, Reverse
- Associate, analogy

Essential Characteristics of a Creative Idea

Ideas of Newell, Shaw and Simon

- Novelty and usefulness (either for the individual or society)
- Rejection of previous ideas
- Results from intense motivation and persistence
- Clarification of vague ideas

Margaret Boden's view

- P-Creativity (Psychological or Individual)
- H-Creativity (Historical or Collective)

How Generative AI Can Augment Human Creativity

- Companies have used crowdsourcing and idea competitions to involve outsiders in the innovation process
- Crowdsourcing, for instance, may produce a flood of ideas, many of which end up being dumped or disregarded because companies have no efficient way to evaluate them or merge incomplete or minor ideas that could prove potent in combination.



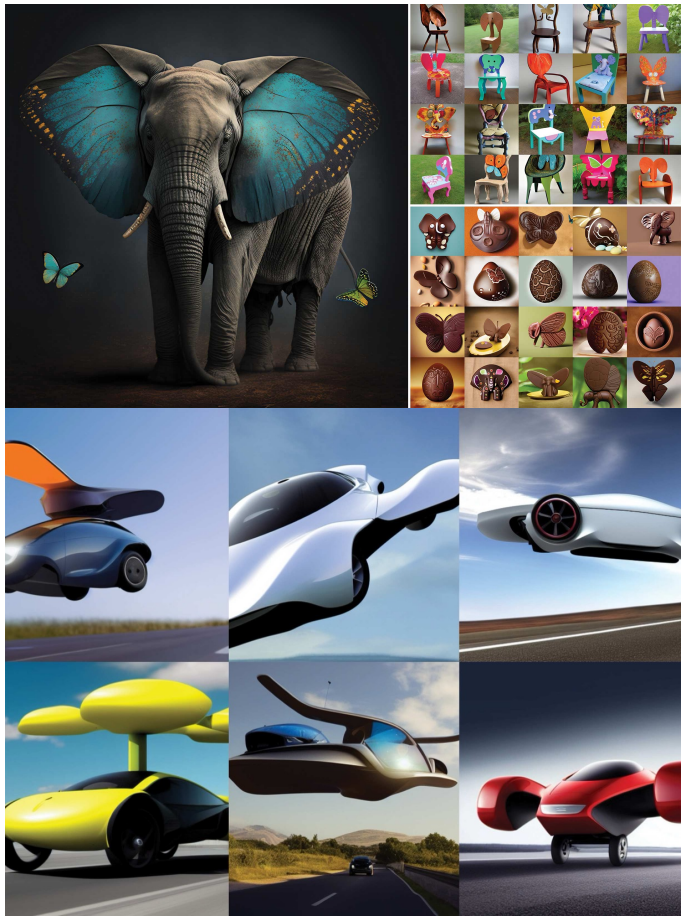
https://hbr.org/2023/07/how-generative-ai-can-augment-human-creativity

- Concept of *Relative Creativity*, where an AI model is deemed as creative as a hypothetical, yet realistic, human creator if it can produce works indistinguishable from that creator, as determined by an evaluator.
- *Statistical Creativity* a means for understanding whether and to what degree AI achieves creativity by comparing it with existing human creators.

https://arxiv.org/pdf/2401.01623

Why AI in creativity?

- Boundary (flexible) Promote Divergent Thinking
- One idea + another: Concept + Theme
- Bias is too biased! Challenge Expertise Bias
- Idea evaluation bottleneck! - Assist in Idea Evaluation
- Improvement and refinement issues



But!

- “It’s easy for AI to come up with something novel just randomly. But it’s very hard to come up with something that is novel and unexpected and useful.”

— John Smith, Manager of Multimedia and Vision at IBM Research

- Can we take what humans think is beautiful and creative and try to put that into an algorithm? I don't think it's going to be possible for quite a while.”

— Jason Toy, CEO, Somatic

- What I'm looking for out of cognitive systems is not just another form of computing but something that actually creates a presence in our life and through that presence is able to inspire us.”

— Rob High, Vice President and CTO, IBM Watson

Research Papers

- https://www.sciencedirect.com/science/article/pii/S0004370298000551
- https://onlinelibrary.wiley.com/doi/full/10.1002/jocb.597
- https://www.researchgate.net/publication/371956576_Using_Artificial_Intelligence_for_enhancing_Human_Creativity/link/649e7ec8b9ed6874a5e9a1f6/download
- https://www.degruyter.com/document/doi/10.1515/9783110198560/html?lang=en
- https://iopscience.iop.org/article/10.1088/1742-6596/1760/1/012032/meta

Creativity

- AI alone
- AI + AI
- AI + Human
- Multi machine – multi-thinkers

Creativity or illusion of creativity?

- Do we need to redefine creativity?
- Do we need tool to differentiate human and machine creativity?
- What could be that tool?

Business – creativity and Machines

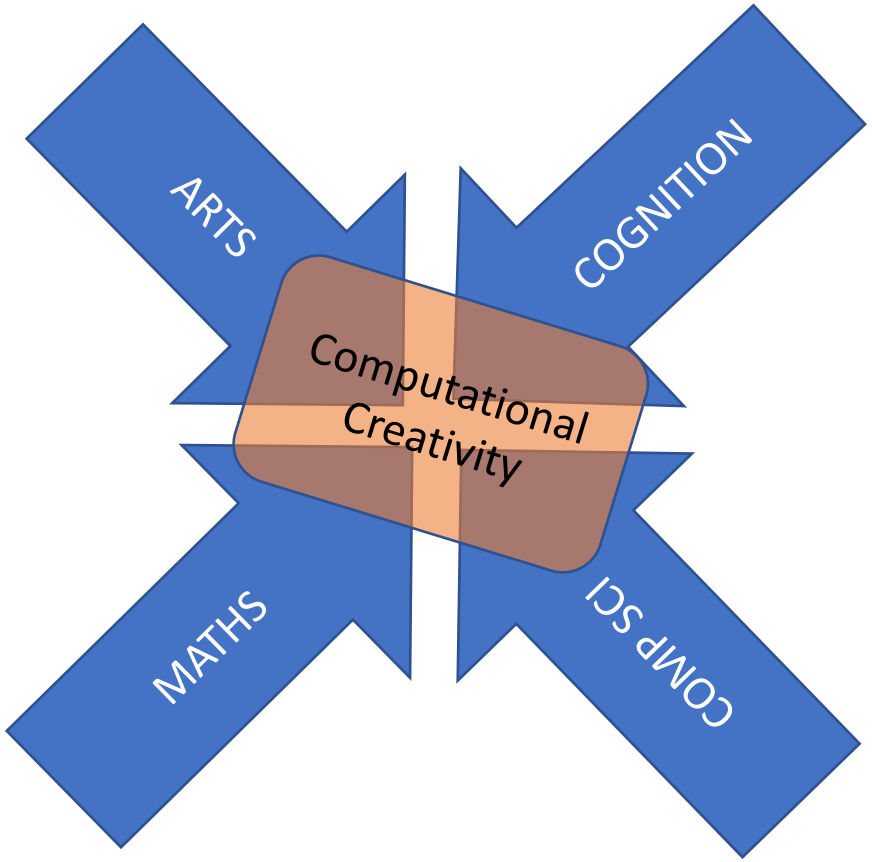
- Synthetic creativity on demand, as currently generated by AI, is certainly a boon to business and marketing. Recent examples include:
- AI-enhanced advertising: Ogilvy Paris used Dall-E to create an AI version of Vermeer's *The Milkmaid* for Nestle yoghurts.
- AI-designed furniture: Kartell, Philippe Starck and Autodesk collaborated with AI to create the first chair designed using AI for sustainable manufacturing.
- AI-augmented fashion styling: Stitch Fix utilised AI to capture personalised visualisations of clothing based on requested customer preferences such as colour, fabric and style.

https://www.weforum.org/agenda/2023/06/ai-cannot-replace-human-creativity/

Quick assignment

- Words those are indicative of creativity

Can you represent creativity mathematically?



Combination of:

- Computational Systems and
- Human creator

Autonomous Vs semi-autonomous

- Is it strong vs weak?

Is creativity?

- Exploration?
- Evolution?
- Change?
- Option Creation?
- Association?
- Combination?
- Knowing ahead of time?
- Surprise / Uncertainty / Anti-Pattern?

Creativity and Business

- Repositioning
- Repurpose
- Expansion
- Combination

The ability to produce or develop original work, theories, techniques, or thoughts. A creative individual typically displays originality, imagination, and expressiveness. Analyses have failed to ascertain why one individual is more creative than another, but creativity does appear to be a very durable trait.

Formalizing the Philosophical Concepts

Exploration: Within a conceptual space

Transformation: Out of the box(space)

Formalization :

- Conceptual Space C a strict subset of set of all concepts U
- Axiom1:Every concept c is a distinct member of U
- Axiom2:Every conceptual space includes F (empty concept)

Formalization (contd…)

- Rules : R(Existence in a space) and T(Transformation in a space) R , T subsets of L
- Interpretation function: [.] and Traversal function: <.>
- Exploratory conceptual space as a tuple : (U,L,R,T,E)
- Beginning of exploratory creative process : <R + T>({F})
- Evaluating concepts (E) : E(C) = value of the conceptual space.

Where is the AI?

- (RACTER, 1984, https://www.poem-generator.org.uk) generates poems, stories, etc.
  - Rules to be broken or adhered
  - Syntax directives
  - Sentences too bizarre at first look
  - Deeper meaning on repeated thought
- Creativity is in the reader's mind
- Sentences become insignificant soon
- Need for more control

Auto-poem generation

- **Template Based Poetry Generation:** Templates of poetry forms are filled with words that suit the defined constraints (either syntactic, rhythmic or both). After using the Poetry Creator and looking at several generated results, it is clear that the system follows this approach.
- **Generate and Test Approaches**: Random word sequences are produced according to formal requirements, that may involve metric, other formal constraints and semantic constraints. Manurung's chart system, the WASP system and the generate and test strategy of Tra-La-Lyrics follow this approach.
- **Evolutionary Approaches**: Poetry generation is based on evolutionary computation. POEVOLVE and Manu- rung's McGonnagall are two systems that follow an evolutionary approach.
- **Case-Based Reasoning Approaches**: Existing poems are retrieved, considering a target message provided by the user, and are then adapted to fit the required content. ASPERA and COLIBRI are examples of this approach

Ref: Gonçalo Oliveira, Hugo. (2009). Automatic generation of poetry: an overview.

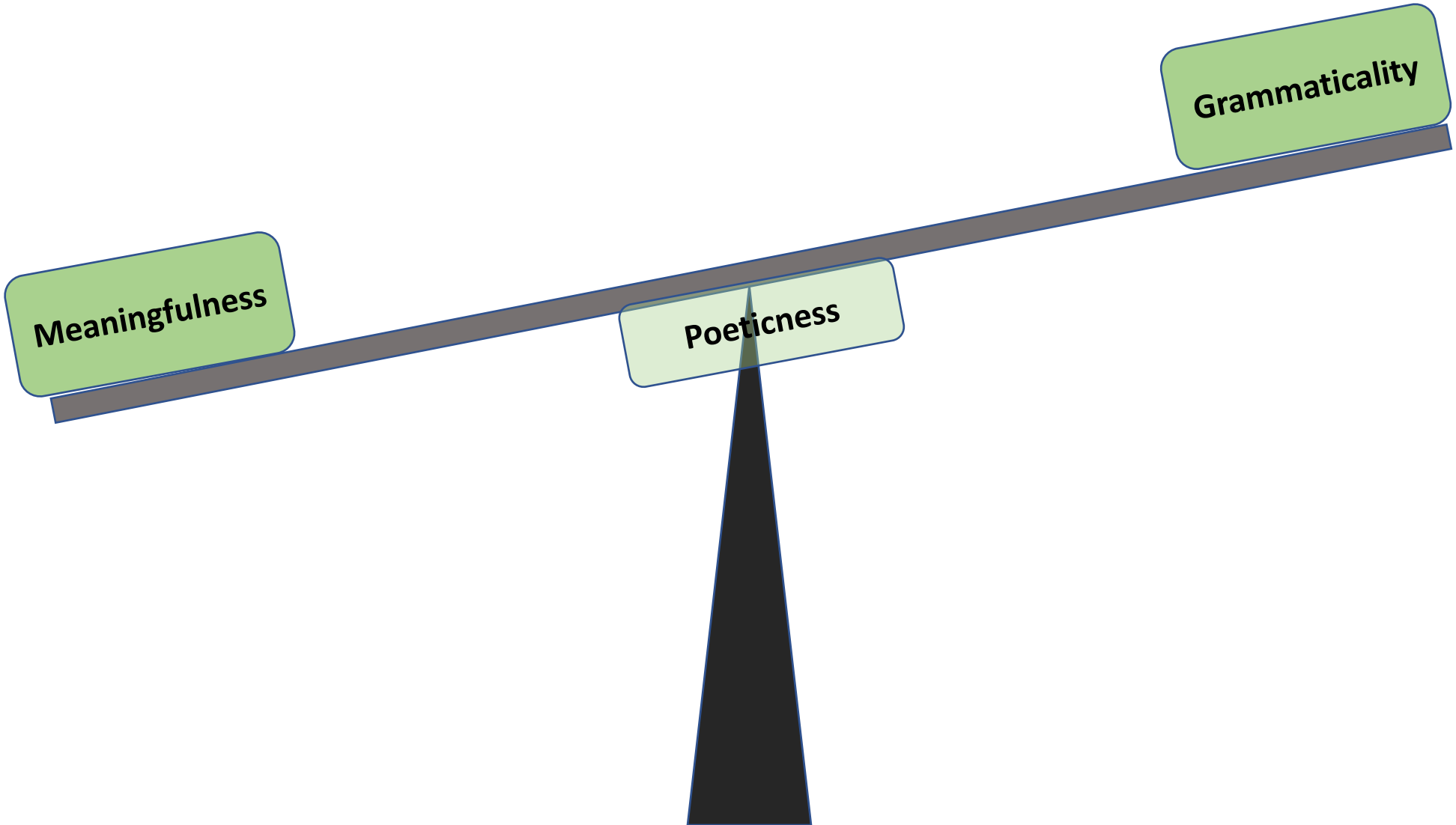
Need for Flexibility

- Rule-based systems are monotonous
- Example: Generating a story (TALE-SPIN)
  - Each object tries to satisfy its goals
  - Creativity is shown only when the plot turns an unexpected way
  - Object need not try to reach goals at every step
- Solution: Use the “meta-” approach
- Develop rules for rules

Rhyme – meaning and art in business

- **Meaningfulness**: convey a conceptual message, which is meaningful under some interpretation.
- **Grammaticality**: obey linguistic conventions prescribed by a given grammar and lexicon.
- **Poeticness**: exhibit poetic features.

Striking the creativity balance!



Examples

- The ALAMO group
- Manurung’s chart system
- POEVOLVE
- WASP
- ASPERA
- McGonnagall

Creativity is the thought process!
Approved by ?

Quick assignment

- Scenario where you thought machine is creative!

Another perspective to Creativity

- A process by which a symbolic domain in culture is changed

Arrange jumbled words

I think do can taught it is simple and thinking be

Possible combinations and which one is creative?

What Can be intelligent? Can they novel and endorse creativity of individuals?

- Products
- Services
- Actions
- Interactions
- Expressions
- Teams

Questions?

- How could you formulate creativity test for humans?
- How can you get rid of subjectivity in creativity? Is it possible?

Study by Avraham Rahimov argues against discarding the Turing Test, proposing instead using more refined versions of it, for example, by interacting simultaneously with both an AI and human candidate to determine who is who, allowing a longer interaction duration, access to the Internet and other AIs, using experienced people as evaluators, etc. The findings highlight that the Turing Test remains an important and effective method for evaluating AI, provided it continues to adapt as AI technology advances.

https://arxiv.org/html/2505.02558v1

What could be the new Turing test?

TURING TEST



Machine Response



Human Responder

Role Change?
Dynamism
Variability
Answer driven
Emotion driven
Human driven



Interrogator / investigator

CREATIVITY TEST



Machine Response



Human Responder

Dynamic Environment



Interrogator / investigator

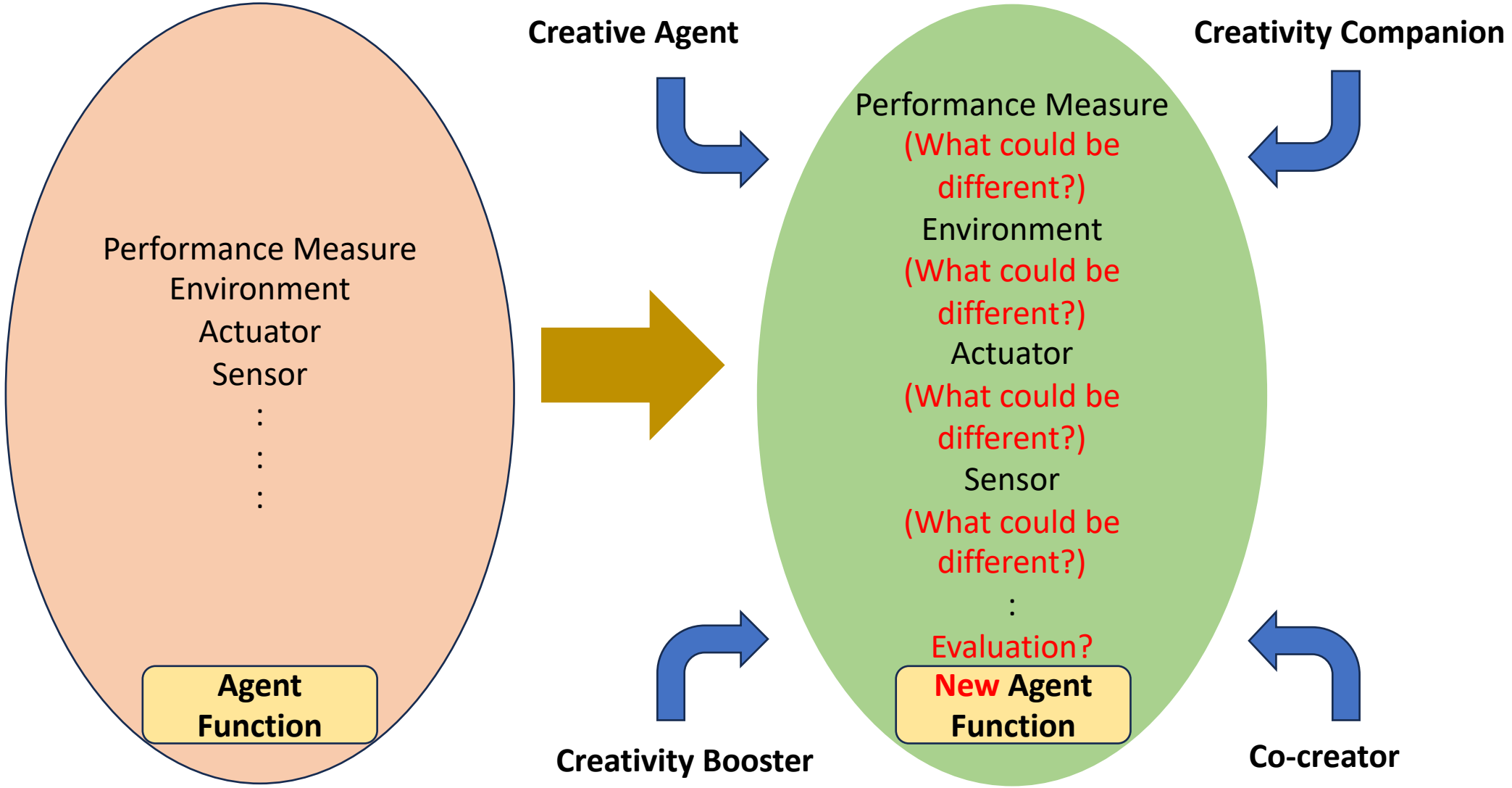
Creativity evaluator

Some responses may be creative!

Cannot detect: Who is the originator of creative response while some of them are also coming from machine

Intelligent Agent

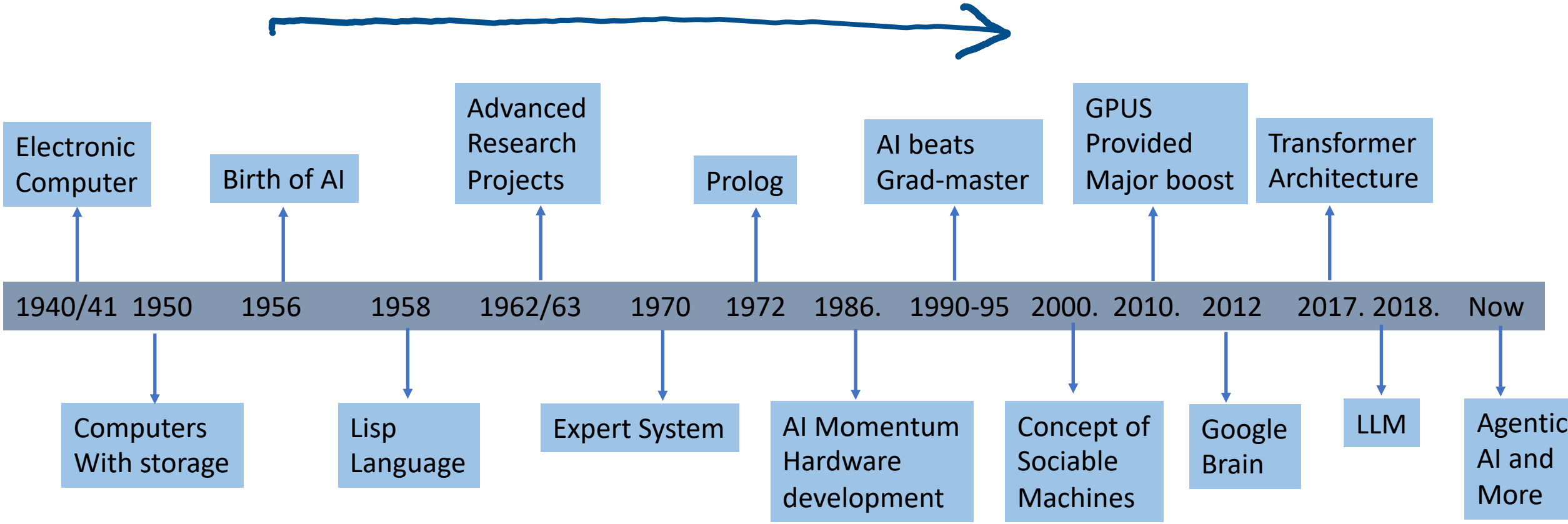
Creative Agent



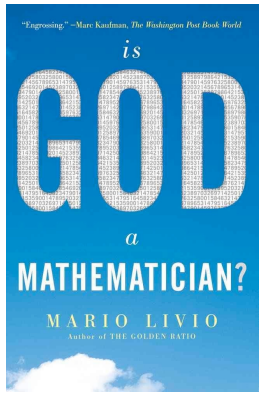
Different approaches and views of creativity:

- Nomothetic versus idiographic view: The nomothetic approach focuses on universals of creativity while the idiographic focuses on individual differences and uniqueness of creative individuals (general laws and patterns that apply to large groups of people)
- Mental versus social phenomenon: Creativity is seen as a mental or psychological phenomenon for the generation of new ideas or as the result of an audience's appreciation of a novel idea.
- Improbabilist versus impossibilist approach: The improbabilist approaches see creativity as a novel combination of familiar ideas while the impossibilists consider creativity as new ideas never appeared before. This is possible thanks to a transformation of the conceptual space.
- Rational versus irrational process: The rational approaches include logic and evolution. Logic sees creativity in processes like abduction (guessing a conclusion given some premises), education – inferring (relate properties of different objects), deduction and induction (relate properties from general to specific and vice versa). Evolutionary approaches instead see creativity as an adaptation to the environment motivated by problem solving while the irrational approach focuses on the emotional part of creativity.

AI Continues…



Thinking…


"Engrossing." —Marc Kaufman, The Washington Post Book World
is
GOD
a
MATHEMATICIAN?
MARIO LIVIO
Author of THE GOLDEN RATIO


EDWARD
DE BONO
TEACHING THINKING

What is thinking? Thinking and AI

- Is it about reasoning?
- Is it about finding relationship?
- Is it about finding process to solve problem
- Thinking is organizing and reorganizing known facts and learning and extrapolating them in the context of problem

Types of Thinking

- Perpetual or concrete
- Conceptual or abstract
- Logical with reasoning
- Creative
- Problem based – problem solving oriented
- Convergent Vs Divergent

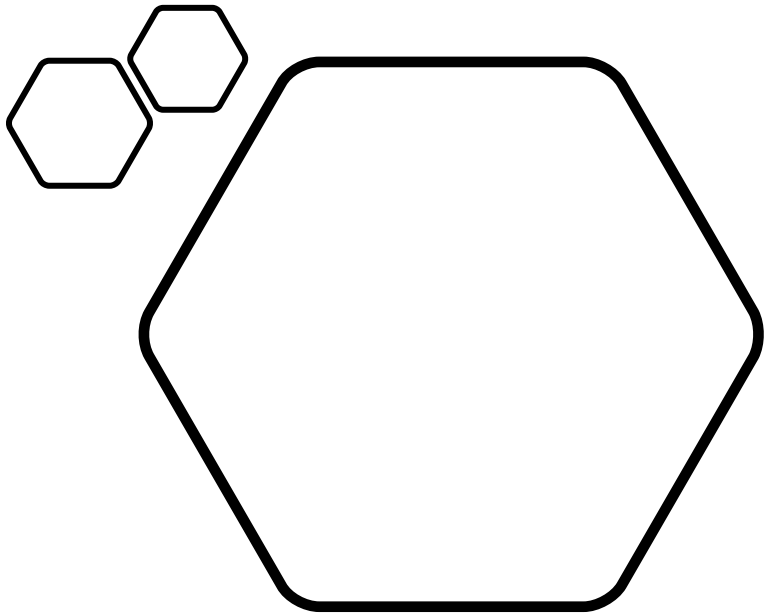
Convergent Vs Divergent thinking

- **Convergent Thinking** focuses on attaining solution of the problem – it aims to converge to solution and hence many times constrained
- In contrast, **divergent thinking** is about looking in different directions – opening mind and going beyond boundaries – it also about trying out multiple solutions for given problem

Convergent Vs Divergent thinking and AI

- Convergent thinking focuses on reaching one well-defined solution to a problem. This type of thinking is best suited for tasks that involve logic as opposed to creativity, such as answering multiple-choice tests or solving a problem where you know there are no other possible solutions.
- Divergent thinking is the opposite of convergent thinking and involves more creativity. With this type of thinking, you can generate ideas and develop multiple solutions to a problem. While divergent thinking often involves brainstorming for many possible answers to a question, the goal is the same as convergent thinking—to arrive at the best solution.
- Conceptual combination, mental flexibility, and depth of semantic knowledge

Thought Experiments – Creative Learning with limited Success



- Taking contrary view
- Thinking one subject in terms of another subject
- Taking multiple perspectives
- Hunting opportunities to think differently
- Resulting new relevant(?) combinations using existing ones

Linking Learning to Creativity:

Learning is building ability to produce the expected results

IDEA Canvas-2: How would you define learning when creativity is in focus?



Finding more through available knowledge through association, meta-reasoning, exploration, expansion, transformation and mapping…

Thinking from Computing and Creative Perspective

- Machine Thinking?
- Cognition, thinking and Computing

Some Ideas we will discuss…

1. Intelligent creative agents – Types and Design
2. Multi-agent systems – with different creative paradigms
3. Cognitive computing and creativity
4. Heuristic search algorithms for creative goals
5. Non-heuristic search algorithms - (what could be goal) - pointers
6. Constraint satisfaction problems – Constrained creativity
7. Expert systems – creativity stored
8. Hybrid intelligent systems
9. Neuro-fuzzy systems
10. AI aspects of mind sharing for creativity – Social networking
11. Creative Connections – connectionist systems
12. Graphical models for Creative AI
13. Immune systems – negative selection of algorithms
14. Collective intelligence
15. Co-operative learning
16. Behavioral system
17. Human competitive evolution-based algorithms
18. Reinforcement Machine Learning – (Creativity reinforcement paradigm)
19. Creative AI products
20. Embedding p-creativity and h-creativity in machines

Pointers with Objectives

CONCEPTS

- What is AI?
- Human Learning,
- Creativity
- Thinking
- Conceptual Spaces
- Experience and Examples
- Predictions
- Intelligence
- Learning models and Paradigms
- Building blocks of Intelligent systems
- Algorithms
- Applications
- Case-studies

APPROACHES

- Statistical
- Associative
- Behaviorism
- Evaluation
- Evolution
- Exploration
- Genetic
- Symmetry
- Analogy
- Combination
- Time-Series Predictive
- Immune System "Group Behavior"
- Discriminative
- Generative

Living Geometry

- The human need for living geometry is supported by interdisciplinary evidence from biology, environmental psychology, and neuroscience.
- Used together with generative AI, living geometry therefore offers a useful method for both creating and evaluating designs based on objective criteria. Adopting a hybrid epistemological frame work of AI plus living geometry as a basis for design uncovers a flaw within contemporary architectural practice.

Ref: Salingaros, N.A. Environments That Boost Creativity: AI-Generated Living Geometry. *Multimodal Technol. Interact.* **2025**, *9*, 38. https://doi.org/10.3390/mti9050038



Ref: Salingaros, N.A. Environments That Boost Creativity: AI-Generated Living Geometry. *Multimodal Technol. Interact.* **2025**, *9*, 38. https://doi.org/10.3390/mti9050038

Alexander’s 15 fundamental properties

1. Levels of scale
2. Strong centers
3. Thick boundaries
4. Alternating repetition
5. Positive space
6. Good shape
7. Local symmetries
8. Deep interlock and ambiguity
9. Contrast
10. Gradients
11. Roughness
12. Echoes
13. The void
14. Simplicity and inner calm
15. Not-separateness

These geometrical properties act synergistically to generate environments that evoke calm, promote cognitive engagement, and reduce stress.

Psychological Creativity and Historical Creativity

- Examples

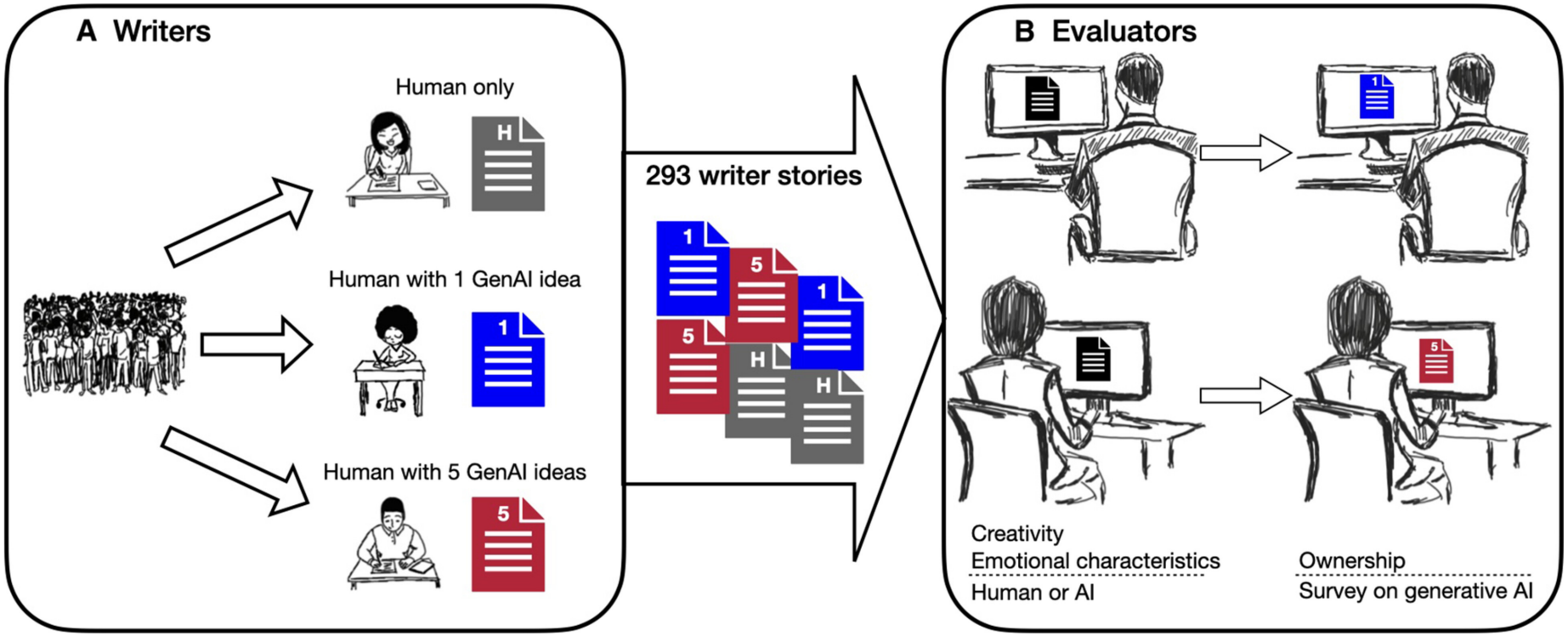




Text-to-image prompt: Cat and horse nervously balancing on a bicycle.

Ref:https://www.adobe.com/products/firefly/discover/how-ai-changes-creative-work.html

AI enhances individual creativity but reduces the collective diversity



Ref: https://www.science.org/doi/10.1126/sciadv.adn5290

Agents for Creativity enhancement with collective diversity inclusion

- Is decentralization in creativity is good idea
- Bad samples could be good sample
- Progress may not have right direction
- Positive + Positive < Negative + Positive
- Samples leading to questions?

Historical Creativity in machines

- A creative task done by machine that – it has not done in the past
- A creative task done by machine that no machine has done in the past

Building Rational Creative Agents

Summary

- Agents
- Applying conceptual spaces to agents
- Applying different creativity concepts to agents
- Defining creativity and new conceptual spaces
- Creativity and thinking

Lecture **Seven, Eight**

Creative Thinking for AI and IoT Innovations

1. Intelligent Agents → Creative Agent: Sense, Learn, Infer and Act (Consciousness) → New actions
2. Mind maps, Creative connections and innovation
3. Creative connections, state transitions, human actions and machine actions for co-creation

Parag Kulkarni, PhD, DSc

Reductionism

- Reductionism sees the parts as paramount and seeks to identify the parts, understand the parts and work up from an understanding of the parts to an understanding of the whole.
- Holism considers systems to be more than the sum of their parts. It is of course interested in the parts and particularly the networks of relationships between the parts, but primarily in terms of how they give rise to and sustain in existence the new entity that is the whole whether it be a river system, an automobile, a philosophical system or a quality system. It is the whole that is seen as important and gives purpose to the study.

Reductionist Creativity, Connectionist Creativity and Inferential creativity

- Reductionist – Part at a time
- Connectionist – Find connections and association
- Inferential – Inference based on relationships

Reductionist and Connectionist Agents

Leonardo da Vinci

To develop a complete mind:

Study the science of art;

Study the art of science.

Learn how to see.

Realize that everything

connects to everything else.

Embedding creativity in machines and rational agents

Definitions

- Consciousness
  - Awareness; free will; the state of being aware of and responsive to one's surrounding
- Epiphenomenon
  - Dictionary: "A secondary phenomenon that results from and accompanies another"
  - Conscious epiphenomenalism: "Mental states are produced by physical states but have no causal role to play"
  - Epiphenomenalism is the philosophical view that mental states (like thoughts and feelings) are caused by physical brain processes but, in turn, have no causal effect on the physical world.

    – Blackmore 2004
  - Physical → Mental but NOT Mental → Physical

Social Self-Awareness

- Three parts of self-awareness
  - Sense of continuity
  - Sense of personal agency
  - Sense of identity
- Self-awareness allows for introspectively based social strategies for competing and cooperating Gallup (1998)
- Conscious decisions do not have a causal relationship with voluntary motor control

Rational Agent Should

- Responsive: Respond in timely fashion to perceived environment.
- Pro-active: Should exhibit opportunistic, goal directed behavior and take the initiative wherever necessary.
- Social: Should be able to interact when they deem appropriate with other artificial agents, humans in order to compete problem solving.

Revisiting PEAS

- PEAS: Performance measure, Environment, Actuators, Sensors
- Must first specify the setting for intelligent agent design
- Consider, e.g., the task of designing an automated taxi driver:
  - Performance measure
  - Environment
  - Actuators
  - Sensors

IDEA Canvas-2: Let us list different agents and see what is PEAS for them



PEAS

- Must first specify the setting for intelligent agent design

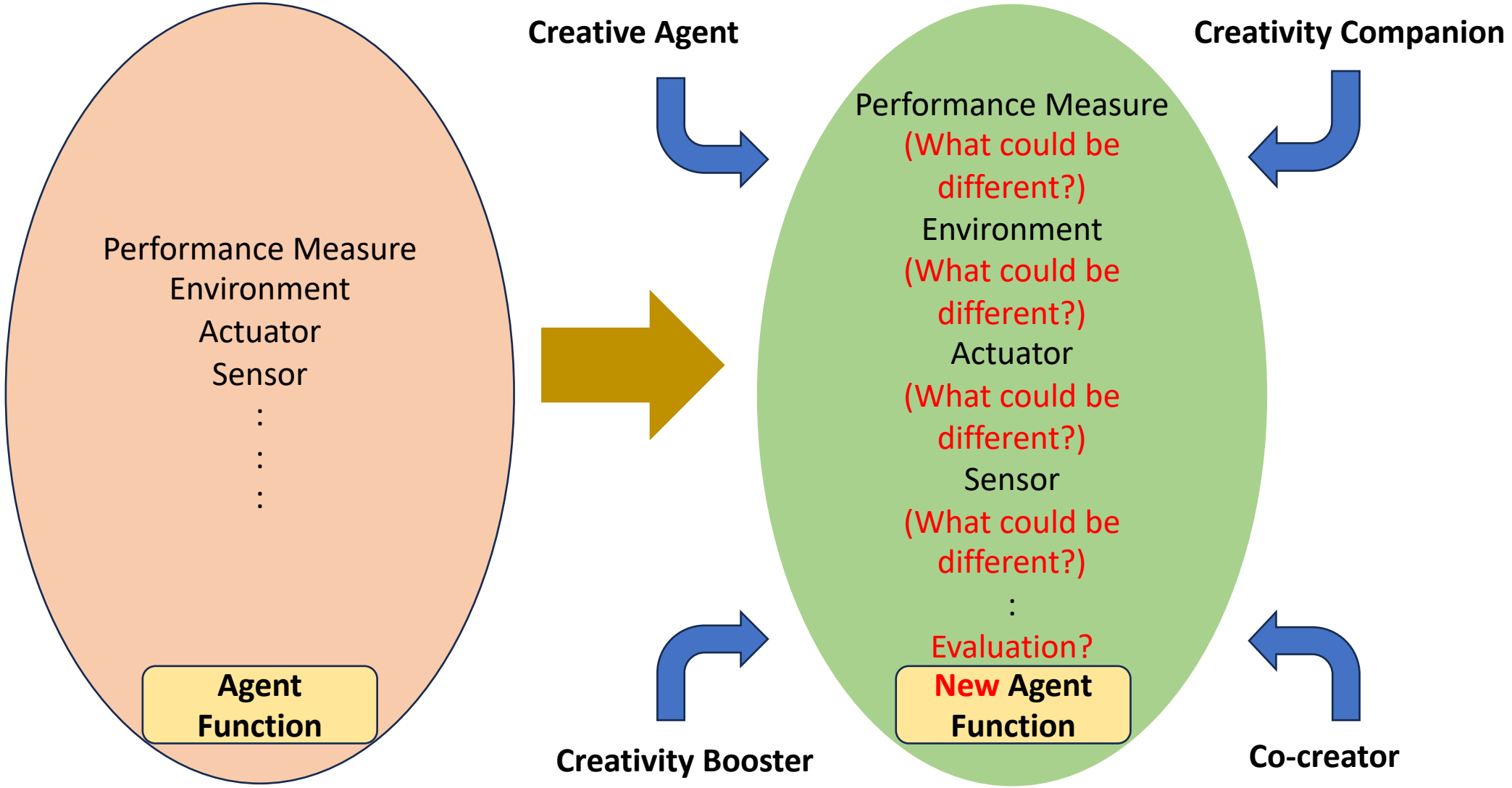
- The PEAS description for it will be:
- **P**: Cleanliness of clothes, time taken, electricity consumed, water used, detergent used, noise made.
- **E**: Clothes, water tap, water, detergent
- **A**: Motor rotating the drum/ spindle, time indicator, operation completion alarm
- **S**: Water level sensor, detergent sensor, cloth weight sensor, time clock.

PEAS

- Medical diagnosis system
- Auto door opening
- Face authentication

Intelligent Agent

Creative Agent



Medical Diagnosis System

P - Accuracy of diagnosis, time taken, additional helpful parameters, recovery time, expenses

E - Patient, doctors, system, equipment

A - Diagnosis decision, further treatments/tests/alerts, medicines

S - BP measuring, Blood- sugar measuring, heart beat track, ECG, EEG, symptoms of patient with regard to the disease, previous reports

Auto door operating System

P - Time taken, efficiency, idle time, response time, noise made

E - Area on both sides of the door

A - Motor closing and opening the door

S - Camera on both sides, object sensing mechanism

Face Authentication System

P - Time taken, accuracy, efficiency, handling ability in case of new face

E - Face

A - Command or message indicating outcome

S – Camera/ Sensor sensing presence

Auto Car System

P - Time taken, accuracy, efficiency, safety

E - Streets, pedestrians, other vehicles, traffic signals

A - Steering wheel, accelerator, brake, horn

S - Cameras, GPS signals, speedometer, engine sensors

Environments

- What are dimensions of environment?
- Fully observable (vs. partially observable): A fully observable environment is the one where the sensors of the agent can detect all aspects that are relevant. In short the agent sensors describe the environment fully or at each point the environment is entirely observable. A crossword puzzle is a fully observable environment, where as in case car system, it is partially observable.
- Deterministic (vs. stochastic): In case of deterministic environment, the next state of the environment is described completely by the current state as well as by the action of the agent. The uncertainty factor is not an issue to be looked at in case of fully observable deterministic environment.
- Discrete (vs. continuous): A discrete environment has a limited / finite number of distinct clearly defined percepts and actions. While in continuous environment those cannot be clearly defined. Chess game has a finite number of discrete states while it is not the case with football.

Environments

- Episodic (vs. sequential): In typical terms the agent's environment is divided into atomic 'episodes'. Each episode or we would call it current scene consists of agent's perception and action. So action choice in an episode wholly depends only on that episode.
- *Static (vs. dynamic):*So, we can say that environment is unchanged while an agent is deliberating. In the sense the entire scenario remains to be as it was when the agent is acting on. Vice versa is the case with dynamic environment, where the environment changes while the agent is taking action. A semi-dynamic environment is the one where the environment does not change with time slab, but the agent's performance does. In real life many examples are dynamic.

Single and multi-agent systems

Environment Types

	Chess with a clock	Chess without a clock	Car driving	Image analysis
Fully observable	Yes	Yes	No	**Yes**
Deterministic	Strategic	Strategic	No	**Yes**
Discrete	Yes	Yes	No	**Yes**
Episodic	No	No	No	**Yes**
Static	Semi	Yes	No	**Semi**
Single agent	**No**	**No**	**No**	**Yes**

- The environment type largely determines the agent design
- The real world is (of course) partially observable, stochastic, sequential, dynamic, continuous, multi-agent

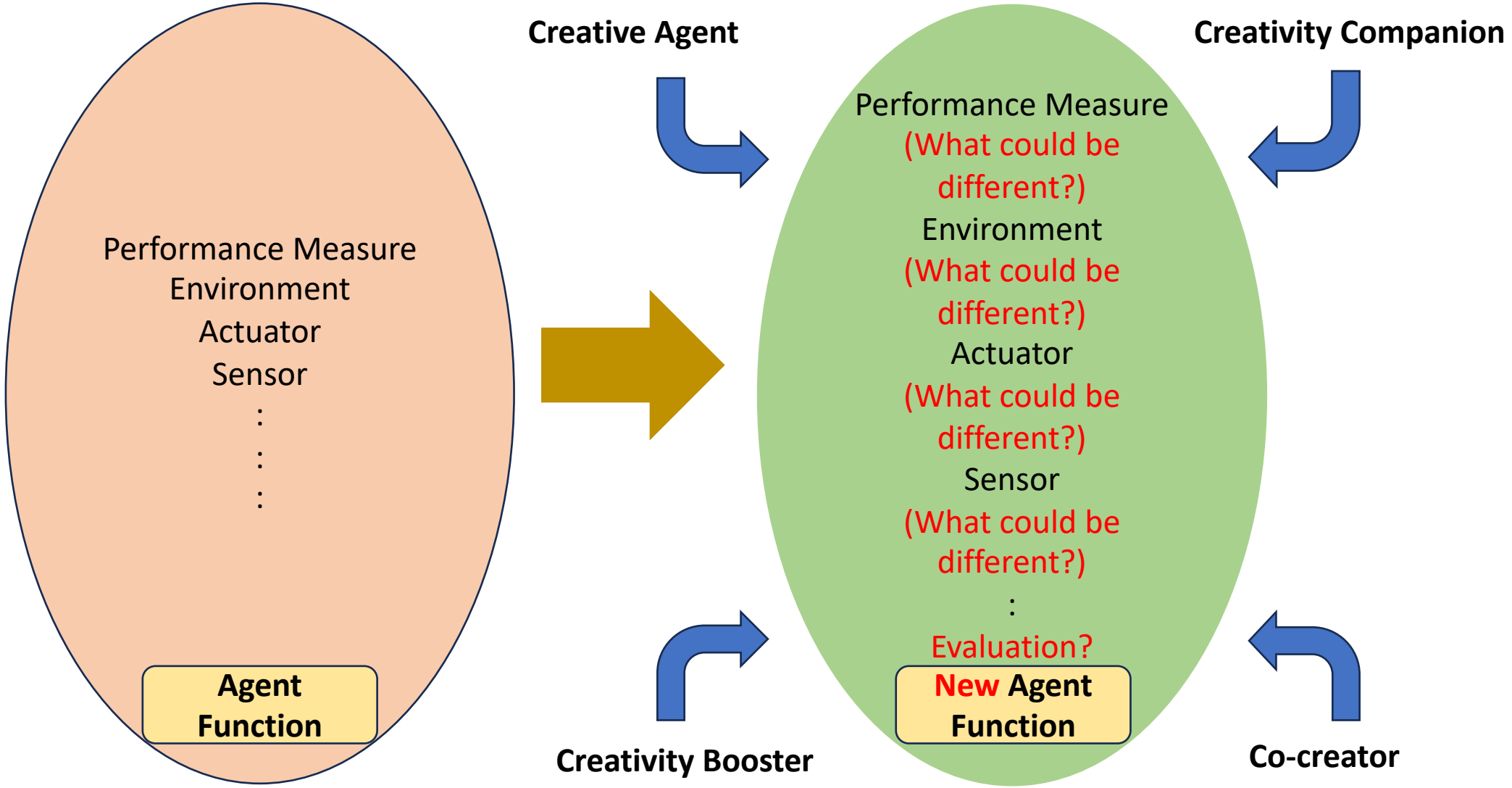
Flexibility and Intelligent Agents

By flexible we mean that system should be:

- Responsive – respond in timely fashion to perceived environment
- Proactive – Should exhibit opportunistic, goal-directed behavior and take the initiative where appropriate
- Social – Able to interact, when they deem appropriate with other artificial agents, humans in order to complete problem solving

Intelligent Agent

Creative Agent



FACTFULNESS POSTER

Dramatic Instincts

Instincts and Creativity

1 The gap instinct



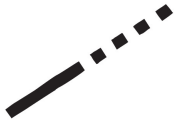
Look at the gap!

2 The Negativity Instinct



It's getting worse!

3 The Straight Line Instinct



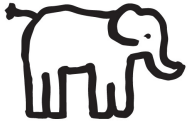
It just continues!

4 The Fear Instinct



It's scary!

5 The Size Instinct



It's big!

6 The Generalization Instinct



They are all the same!

7 The Destiny Instinct



It never changes!

8 The Single Perspective Instinct



This is the solution!

9 The Blame Instinct



That's the bad guy!

10 The Urgency Instinct



It's now or never!

Creative Agent Types!

- Creativity
- Psychological Creativity
- Psychological creativity in machines
- Historical Creativity
- Historical Creativity in machines

How creative agent is expected to learn!

- Deviation from historically established routes
- New variation
- Learn through combination / exploration / expansion
- Dynamic learn set
- Learn set evolves – Algorithm evolves – outcome evolves
- In built Creativity calibrator
- Any ideas?

IDEA Canvas: Social Creative Agents



Other Properties Intelligent Agent Should Have

- Mobility
- Veracity
- Benevolence – avoid conflict – do what is told
- Rationality
- learning

Structure of Agents

- Agent = Architecture + Program

Sensors (current percept) → Agent Programs → Actuators

- An agent is completely specified by the agent function mapping percept sequences to actions
- One agent function (or a small equivalence class) is rational
- Aim: find a way to implement the rational agent function concisely

Agent Environment Relationship

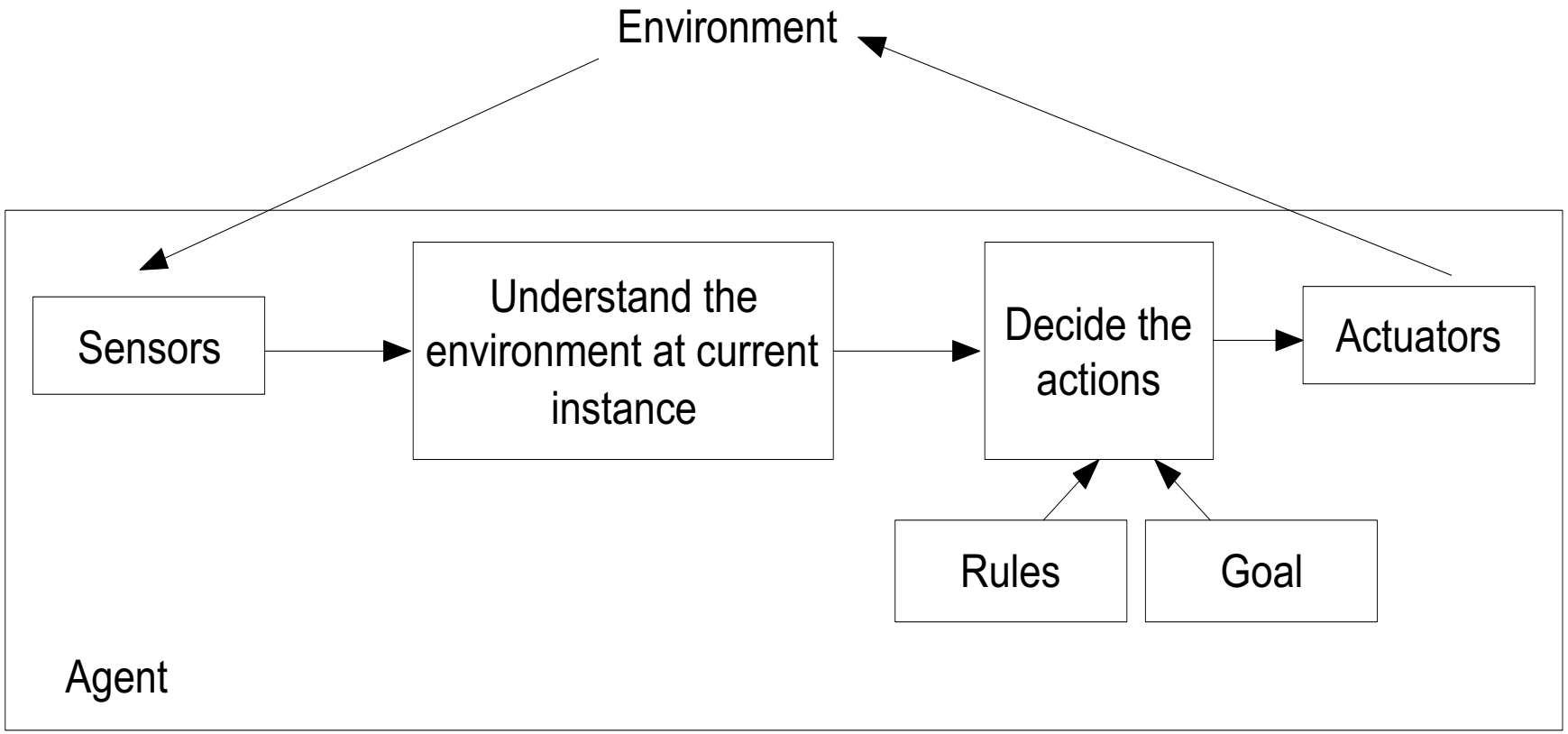


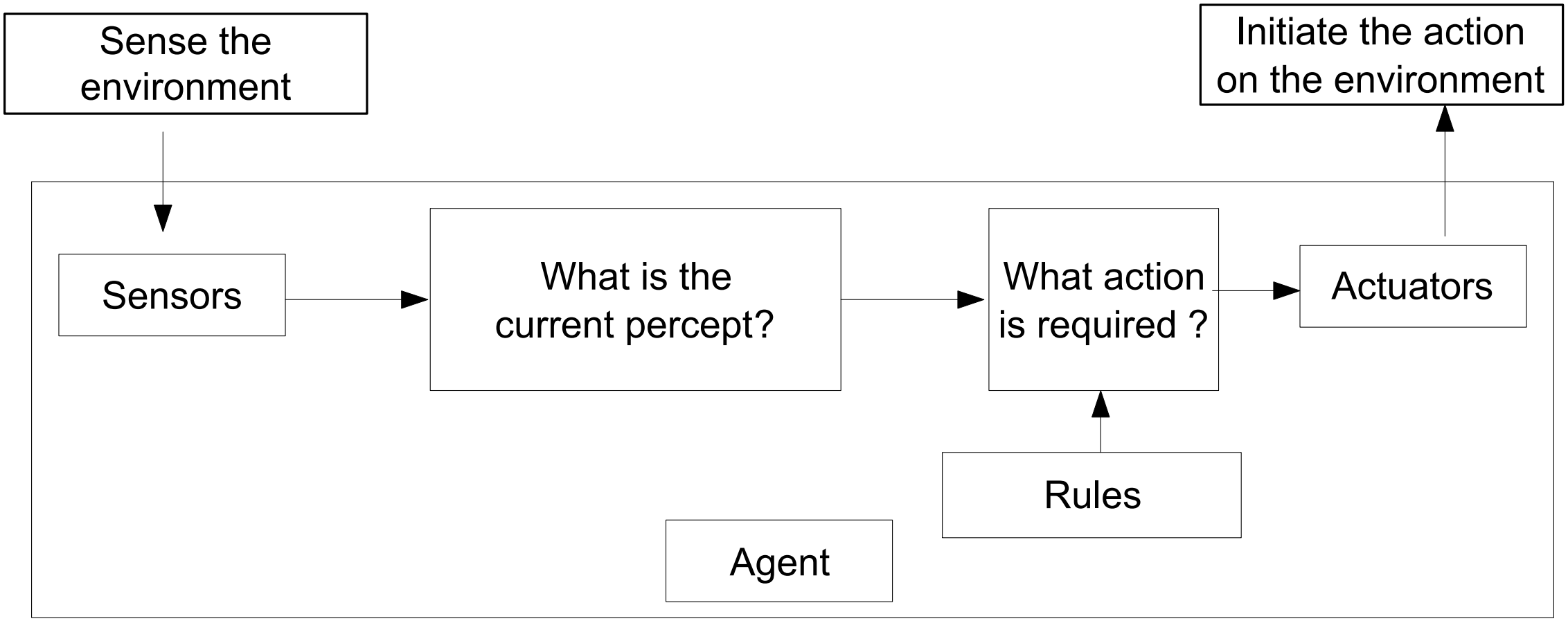
Table Driven Agents

- A table is maintained for decision-making
- Drawbacks:
  - Huge table
  - Take a long time to build the table
  - No autonomy
  - Even with learning, need a long time to learn the table entries

Agent Types

- Simple reflex agents
- Model-based reflex agents
- Goal-based agents
- Utility-based agents

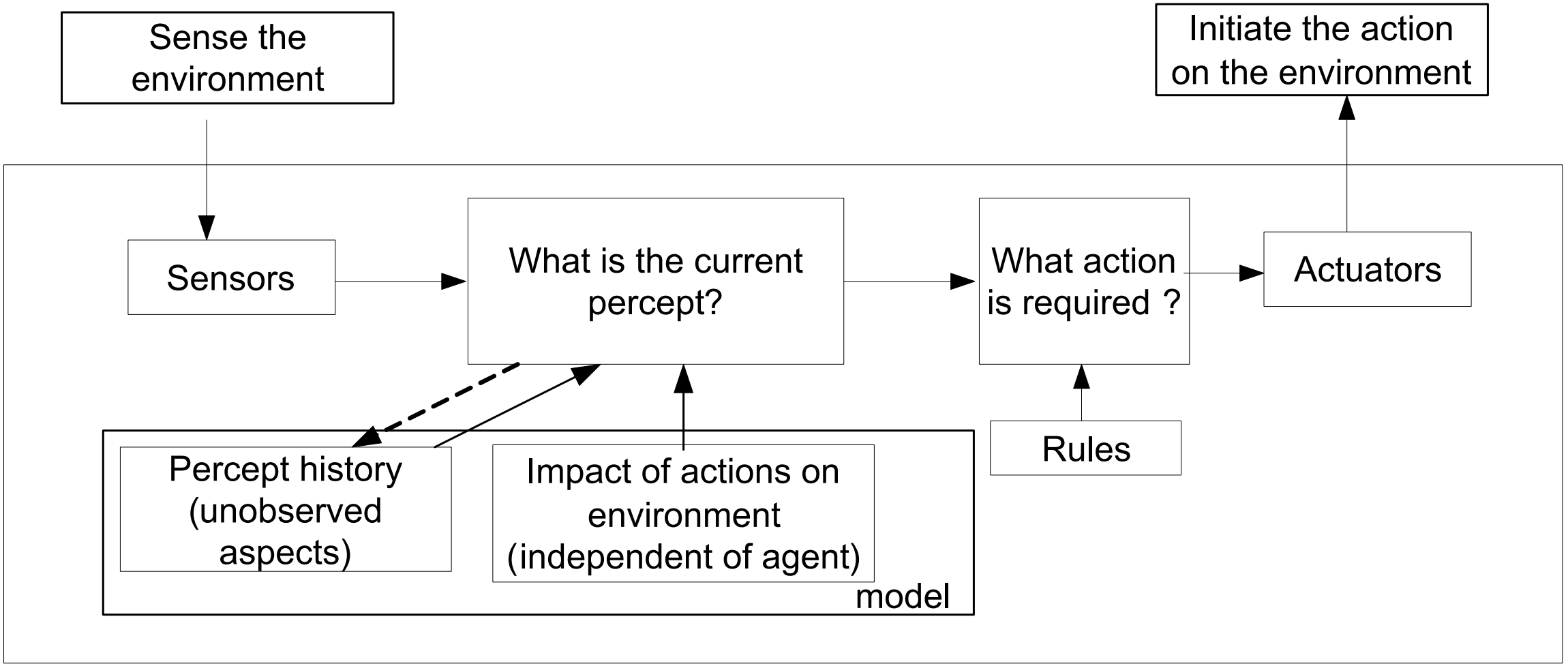
Simple Reflex Agent



Simple Reflex Agent

- Select action on the basis of current percept (ignoring the percept history and not depend on location)
- Condition – action – rule (action driven by condition)
- Simple with limited intelligence
- Useful only in fully observable environment
- Use of randomize in case of vacuum agent

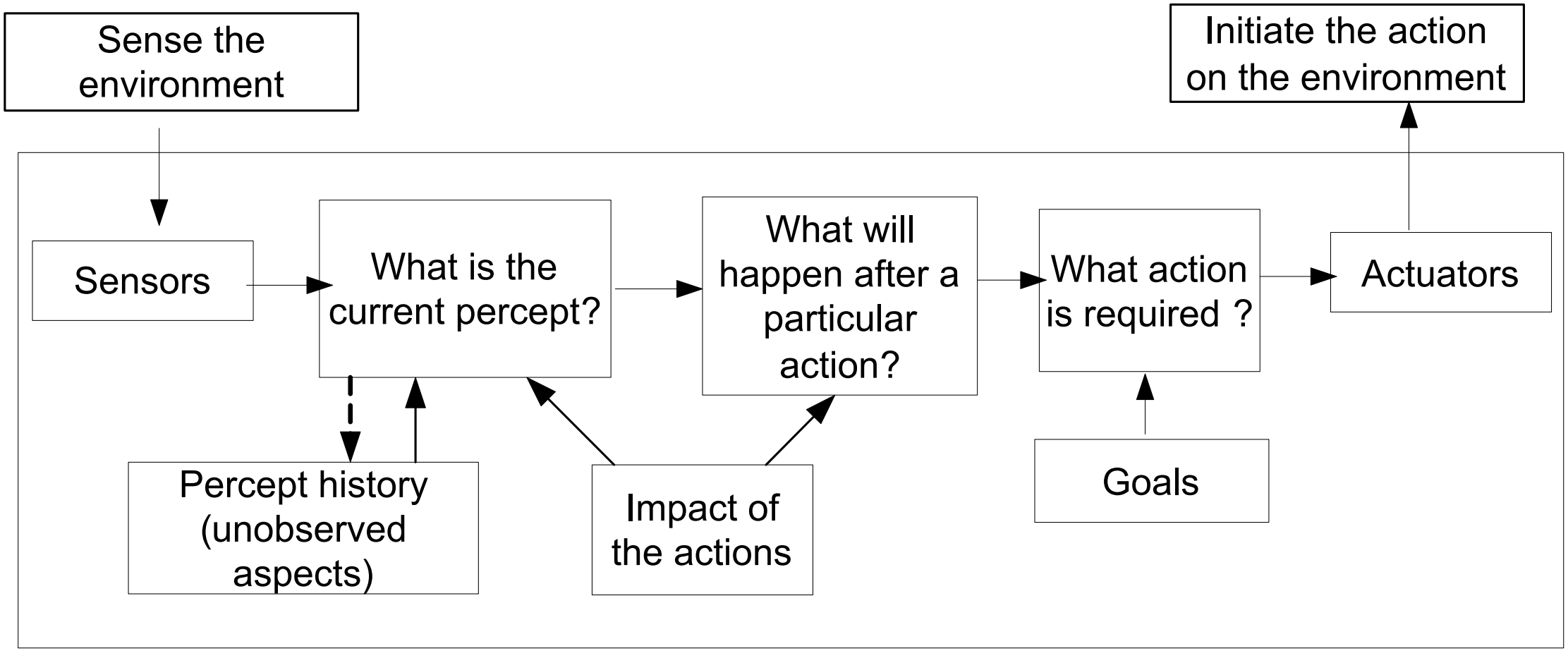
Model Based Reflex Agent



Model Based Reflex Agent

- Keeping track of the world agent can see
- Some sort of internal state – to deal with some un observed aspects
- E.g. previous frame of camera, etc

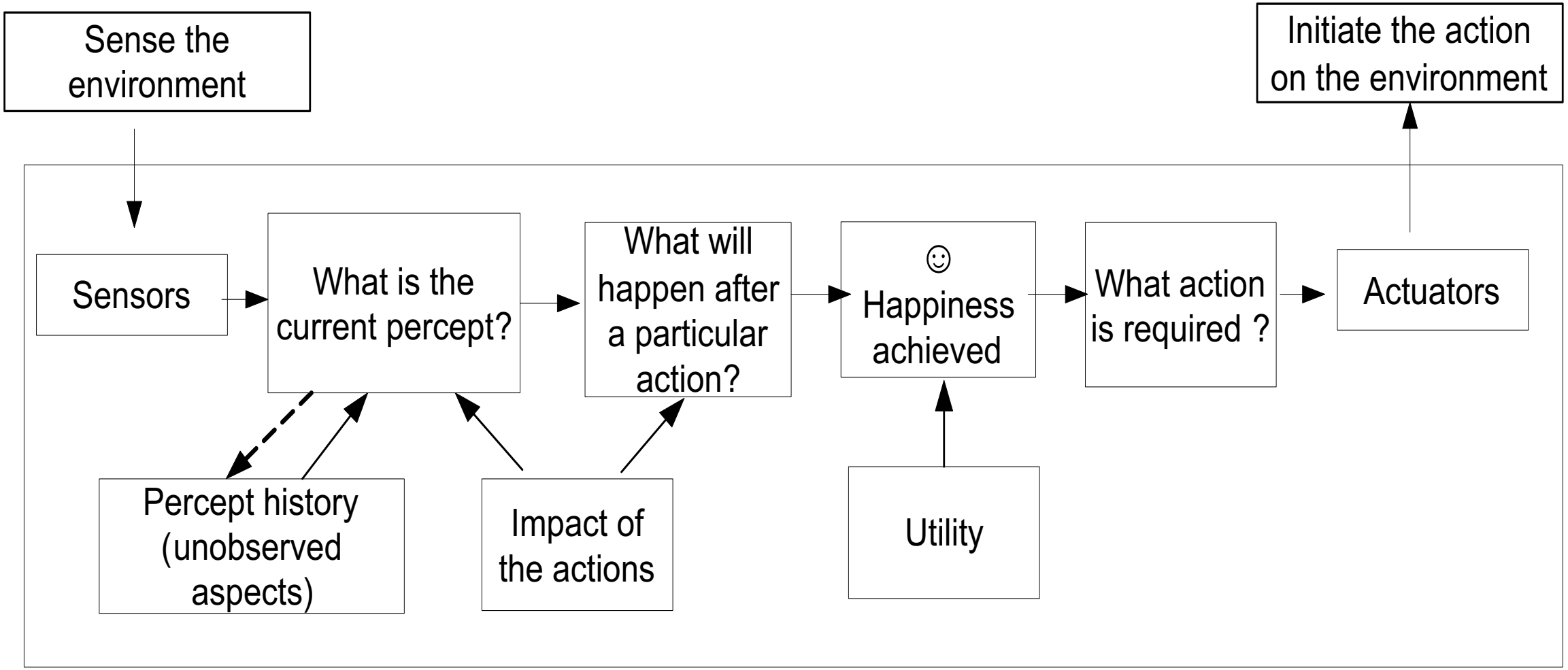
Goal-based Agents



Goal Based Agents

- Apart from the current state there is need to know the goal
- Search and planning may require for driving to goal
- More flexible as knowledge supports actions

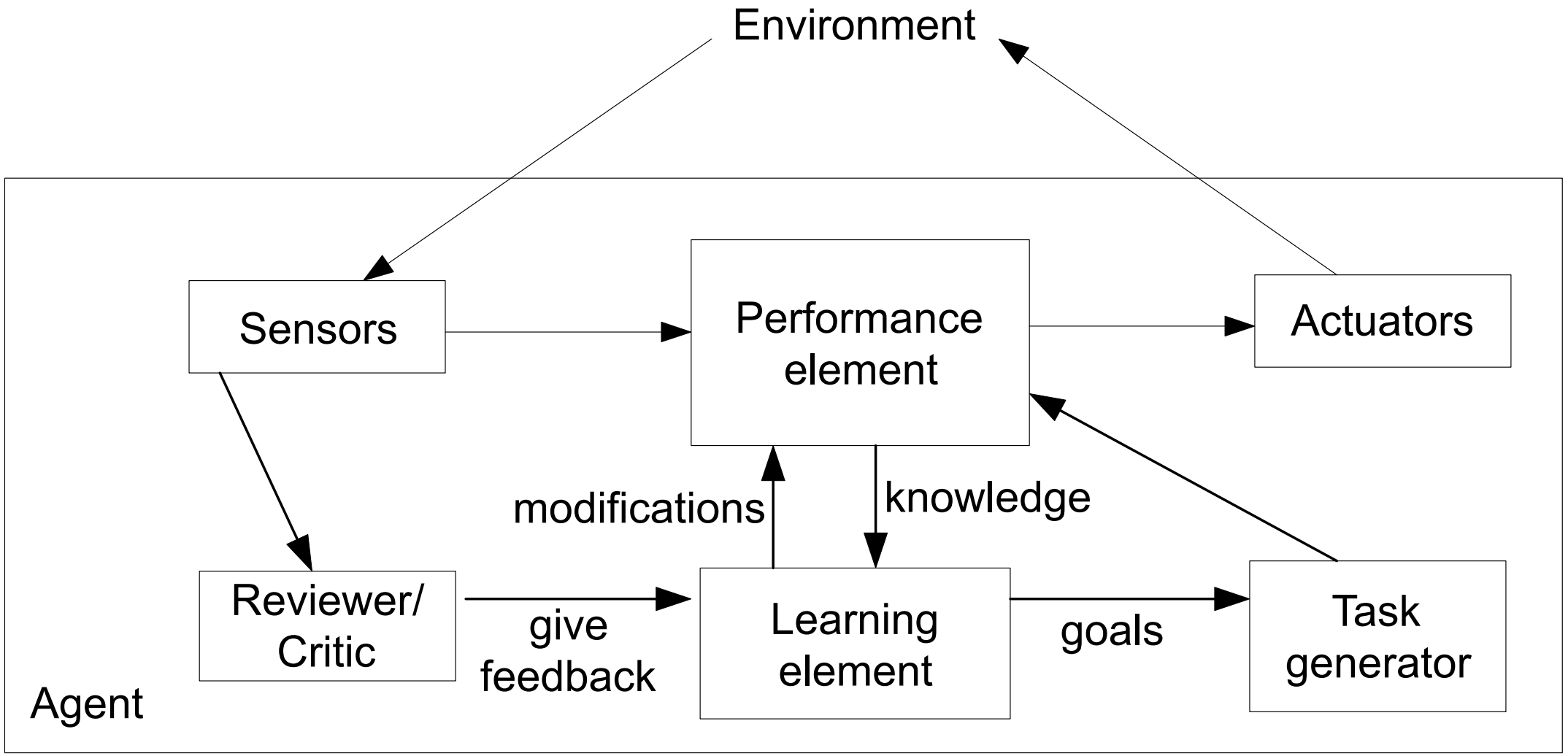
Utility Based Agents



Utility Based Agents

- Not only goal but means are important
- Some routes or series of actions are more safer, quicker, more reliable
- Happy states and unhappy states
- Utility function maps a state on to a real number which describe associated degree of happiness

Learning Agents



Learning Agents

- Ability to learn
- Learning element – responsible for making improvements
- Performance element – responsible for selecting external actions
- Learning agent uses feedback from critic
- Problem generator – responsible for suggesting actions that will lead to new and informative experiences
- Observation of behavior as a result of actions

Learning Agents

- Rewards and penalties
- Learning in an intelligent agent: process of modification of each component of agent to bring the components in to closer agreement with available feedback information – thereby improving the overall performance of the agent.

Other Systems

- Multi-agent systems
- Applications to solve problems those ate not solved so far – or solve already solved problems in better way

Creative Agent

- Function that can measure extent of creativity
- Has ability to perform obvious tasks and creative task
- Rational but non-obvious
- What could be structure of creative agent
- What type of AI problems creative agent is expected to address
- Classify tasks and outcomes on creativity scale
- Sensing previous performance – sensing competing performance

Applications

- Open systems: dynamically changing systems
- Complex systems: Abstraction
- Process control
- Manufacturing
- Traffic control
- Information management
- E-commerce
- Business process management
- Medical domain – monitoring, etc
- Games

Limitations Of Agent Solutions

- No overall system controllers
- No global perspective

Agent Development Bottlenecks

- Requirement specifications
- System design
- System implementation
- System testing

Building Interactive Simulation Environment

- Automated pilot
- Can possible in other domains like entertainment, manufacturing and training

Think About

- New agent type that can enhance agent understanding of evolution of human intelligence
- Buyer agents and seller agents in E-business
- Intelligent agents in network management
- Intelligent agents in Business Intelligence

Talking about uninformed and informed search strategies

Summary

- Learning Agent
- Learning Opportunities
- Adaptive learning
- Different types of learning
- Objectives
- Thinking
- Computing

Thanks